

A Simple Transformation Approach to Difference-in-Differences Estimation for Panel Data

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June 8, 2026

Abstract

In the case of panel data with staggered interventions, we propose a simple time-series transformation that can be combined with various treatment effect estimators. The transformation is at the unit level, and simply requires computing the average outcome prior to an intervention, subtracting it from a post-treatment outcome, and then carefully selecting the control units in each time period. We show that under no anticipation and parallel trends assumptions, the cohort treatment indicators satisfy the key unconfoundedness assumption with respect to the transformed potential outcome. Given identification, any number of treatment effect estimators, including matching estimators, can be applied for each treated cohort and calendar time pair where the average treatment effects on the treated are identified. The doubly robust method of combining inverse probability weighting with linear regression adjustment works particularly well in terms of bias and efficiency. Importantly, our transformation is easily modified to account for unit-specific trends, allowing relaxation of the parallel trends assumption even after conditioning on covariates.

Keywords: Difference-in-differences; Doubly robust estimators; Heterogeneous trends; Panel data; Parallel trends

1 Introduction

The two-way fixed effects (TWFE) estimator, applied to a linear panel model with a constant treatment effect, has been regularly applied in difference-in-differences settings. The estimator is easy to understand and is taught in courses that cover panel data methods. Recently, several authors have pointed out shortcomings of the constant effect model under staggered interventions, including [Borusyak, Jaravel and Spiess \(2024\)](#), [Goodman-Bacon \(2021\)](#), and [de Chaisemartin and d’Haultfoeuille \(2020\)](#). These papers derive alternative representations of the simple TWFE estimator when the treatment effects are heterogeneous across treatment cohort or calendar time.

Other authors have proposed more flexible estimation methods that uncover average treatment effects on the treated in the staggered intervention case. These include [Callaway and Sant’Anna \(2021\)](#) [[CS \(2021\)](#)], who propose long-differencing strategies and apply standard treatment effect estimators. [Sun and Abraham \(2021\)](#) [[SA \(2021\)](#)] propose a fixed effects estimator applied to a more flexible model. Both [SA \(2021\)](#) and [CS \(2021\)](#) are event-study, or “leads and lags” estimators that use only the single period prior to the first intervention time as the reference period. As recently discussed in [Wooldridge \(2025b\)](#), when the [SA \(2021\)](#) approach is extended to allow covariates to appear in a fully flexible way, the resulting estimator is the same as the [CS \(2021\)](#) pairwise regression adjustment estimators using the never-treated group as the control group. The label “leads and lags” comes from the fact that the regression includes the lead values of the treatment indicators in addition to the lags. [Wooldridge \(2025b\)](#) contrasts the leads and lags approach with the “lags only” approach, which omits all pre-treatment indicators and, in effect, averages all pre-treatment periods to construct a reference period. In both cases, one includes a full set of interactions between the covariates and any of the dummy variables included in the regression. [Wooldridge \(2025b\)](#) also shows the pooled ordinary least squares (POLS)

estimator is equivalent to the TWFE estimator on the same equation, but where the time-constant cohort dummies and time dummies are swept away by the TWFE transformation. [Borusyak, Jaravel and Spiess \(2024\)](#) propose imputation estimators based on pooled OLS regressions, where the first includes unit and time fixed effects. [Wooldridge \(2025b\)](#) shows that, with time constant covariates and a balanced panel, the imputation estimates are identical to the POLS (and therefore TWFE) estimates in the flexible model using the entire sample.

An attractive feature of the CS ([2021](#)) approach, which builds on [Abadie \(2005\)](#) and [Sant’Anna and Zhao \(2020\)](#) for the two-period case, is that it permits the application of treatment effects estimators beyond regression adjustment. However, as mentioned above, the CS ([2021](#)) method uses only the period just prior to the intervention in defining a reference period, thereby discarding potentially useful information in earlier time periods. [Wooldridge \(2025b\)](#) shows that, under the standard “error components” structure, with a homoskedastic time-constant component and homoskedastic and serially uncorrelated idiosyncratic errors, the lags only POLS estimator is both best linear unbiased (BLUE) and asymptotically efficient. These results imply that the CS ([2021](#)) estimators are less efficient under a standard set of assumptions. On the other hand, under strong forms of positive serial correlation, CS ([2021](#)) can be more efficient because it is similar to a first-differencing estimator. See [Harmon \(2024\)](#), [de Chaisemartin and d’Haultfoeuille \(2023\)](#), and [Wooldridge \(2025b\)](#) for further discussion.

In this paper, we propose an alternative “rolling” approach that uses information on all pre-treatment outcomes while permitting application of a variety of treatment effects estimators, including regression adjustment (RA), inverse probability weighting (IPW), and the doubly robust IPWRA proposed by [Wooldridge \(2007\)](#). If one prefers, matching on covariates or the propensity score can be used instead. Just as the CS ([2021](#)) can be viewed as applying treatment effects estimators in place of regression in the leads and lags

case, our approach can be viewed as extending the lags only regression methods to various treatment effect estimators.

Another contribution of this paper is that we show how a slight modification of our identification argument establishes identification when (long) differencing is used, as in CS (2021). Our identification argument in Section 4 is not tied to a particular estimation method, and so we extend the CS (2021) class of estimators to include IPWRA estimators, matching estimators, and even causal machine learning estimators. This extension allows one to make direct comparisons between our rolling approach, which uses all pre-treatment periods, and the CS (2021) approach, which uses only the single period before the intervention. We view the approaches as complementary in much the same way that the lags only and leads and lags regression-based estimators described in Wooldridge (2025b) are complementary. Both approaches yield consistent estimators under no anticipation and a popular (conditional) parallel trends (PT) assumption, and they generally have different biases when PT fails. Moreover, neither estimator is guaranteed to be more efficient than the other across empirically relevant settings. Because we show that the same estimation methods, including doubly robust estimators, can be applied to either transformation, it makes sense to try both in empirical research.

In the case of common timing, we show that applying regression adjustment to our transformed outcome variable is equivalent to the regression adjustment estimator based on levels. As mentioned above, Wooldridge (2025b) shows that this estimator is both BLUE and asymptotically efficient under standard assumptions. This provides motivation for applying estimators other than regression adjustment to the transformed variables in order to check robustness of findings. In the case of staggered entry, our approach identifies the average treatment effects on the treated (ATTs) by cohort and calendar time under the same no anticipation and parallel trends assumptions in Wooldridge (2025b). Once identification is established, various estimation methods can be applied.

The remainder of the paper is organized as follows. Section 2 begins with the common timing case, defining the potential outcomes and parameters of interest, and establishing identification assumptions. We start with this case for both pedagogical and theoretical reasons. It is easiest to see the workings of our approach, and to derive certain equivalences, in the common timing case. Moreover, the common timing setting allows straightforward efficiency comparisons with competing approaches. In Section 3 we propose a general approach to estimation using a transformed outcome variable. In Section 4 we extend the framework and identification argument to the staggered case. Section 4.4 discusses how one might accommodate suspected failures of no anticipation, and how one modifies the procedure for unbalanced panels. In Section 5, we show how we can account for heterogeneous trends, focusing on linear trends, to allow violation of the parallel trends assumption (even after we condition on covariates) – a possibility not allowed in CS (2021). In Section 6, we revisit the literature on the effects of Walmart openings on local labor markets. Section 7 contains some concluding remarks. Supplementary Appendix C presents Monte Carlo simulations demonstrating that the rolling methods perform well in terms of both bias and precision.

2 Setup and Identification with Common Timing

We begin with the case where the date of the intervention is the same for all treated units, and the intervention is in place through the final period. The time periods in the population are $t = 1, 2, \dots, T$ and the date of the intervention is S , where $1 < S \leq T$; we assume at least one pre-treatment period. The identification arguments are based on an underlying population, and for each unit i , we use $\{Y_{it}(d) : d \in \{0, 1\}, t = 1, \dots, T\}$ to denote the time series of potential outcomes, where $d = 0$ denotes the control state and $d = 1$ denotes the treated state. Depending on the context, we suppress the unit index i and write $Y_t(d)$ in place of $Y_{it}(d)$ for notational simplicity.

The binary time-constant treatment indicator is D , where $D = 1$ means treatment starting in period S and lasting through period T . A time-varying treatment indicator is $W_t = D \cdot p_t$, where p_t is a post-treatment indicator equal to 1 if $t \geq S$ and otherwise zero.

Like the recent literature on DiD, our focus is on the average treatment effect on the treated (ATT or ATET) in each treated period $r = S, \dots, T$:

$$\tau_r = E[Y_r(1) - Y_r(0) \mid D = 1]. \quad (2.1)$$

The fundamental problem of identification of τ_r is that we only observe the treatment status, D , and the outcome,

$$Y_r = (1 - D) \cdot Y_r(0) + D \cdot Y_r(1). \quad (2.2)$$

Importantly, when $D = 1$, $Y_r = Y_r(1)$, which means

$$E[Y_r(1) \mid D = 1] = E(Y_r \mid D = 1), \quad r = S, \dots, T. \quad (2.3)$$

The expectation $E(Y_r \mid D = 1)$ can be estimated in a consistent, even unbiased, way under various sampling schemes. Under random sampling, the average of Y_r across the treated subsample is unbiased and consistent. Therefore, writing

$$\tau_r = E[Y_r(1) \mid D = 1] - E[Y_r(0) \mid D = 1] = E(Y_r \mid D = 1) - E[Y_r(0) \mid D = 1],$$

it is easily seen that the challenge is in identifying $E[Y_r(0) \mid D = 1]$.

If the treatment is randomly assigned then estimation of $E[Y_r(0) \mid D = 1]$ is easy because, under randomization,

$$E[Y_r(0) \mid D = 1] = E[Y_r(0) \mid D = 0] = E(Y_r \mid D = 0).$$

For most applications, random assignment is too strong an assumption. To see how to

relax it, use simple algebra to write

$$\begin{aligned}
Y_r(1) - Y_r(0) &= \left[Y_r(1) - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_q(1) \right] - \left[Y_r(0) - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_q(0) \right] \\
&\quad + \frac{1}{(S-1)} \sum_{q=1}^{S-1} [Y_q(1) - Y_q(0)] \\
&\equiv \dot{Y}_r(1) - \dot{Y}_r(0) + \frac{1}{(S-1)} \sum_{q=1}^{S-1} [Y_q(1) - Y_q(0)]
\end{aligned} \tag{2.4}$$

where

$$\dot{Y}_r(1) \equiv Y_r(1) - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_q(1) \tag{2.5}$$

and similarly for $\dot{Y}_r(0)$. Note that for each $r \in \{S, S+1, \dots, T\}$, $\dot{Y}_r(1)$ denotes the treated potential outcome in period r , after subtracting the average of the pre-treatment outcomes. The third term is the average of the difference of the pre-treatment period “treatment” effects.

Given the representation in (2.4), we can write

$$\tau_r = E[\dot{Y}_r(1) \mid D = 1] - E[\dot{Y}_r(0) \mid D = 1] + \frac{1}{(S-1)} \sum_{q=1}^{S-1} E[Y_q(1) - Y_q(0) \mid D = 1] \tag{2.6}$$

The first assumption, a “no anticipation” condition, eliminates the third term in (2.6).

Assumption NAC (No Anticipation, Common Timing): For the eventually treated indicator D ,

$$E[Y_t(1) - Y_t(0) \mid D = 1] = 0, \quad t = 1, \dots, S-1. \square \tag{2.7}$$

The name of this assumption derives from the fact that $E[Y_t(1) - Y_t(0) \mid D = 1]$ for $t < S$ are average treatment effects on the treated prior to the intervention, and the assumption is that these are all zero. Assumption NAC is implied by an assumption commonly used in the literature, namely, $Y_t(1) = Y_t(0)$, $t = 1, \dots, S-1$. This assumption is implicit in Heckman, Ichimura and Todd (1997) and made explicit in Abadie (2005) and elsewhere. Because the variable indexing $Y_t(\cdot)$ is treatment status not yet assigned, the assumption

rules out anticipatory changes in the potential outcomes, on average. If one is concerned that the announcement of a policy prior to its implementation may result in some units strategically manipulating their pre-intervention outcomes, one might drop a period or two just prior to the intervention.

Given Assumption NAC, we can express τ_r as

$$\tau_r = E \left[\dot{Y}_r(1) \mid D = 1 \right] - E \left[\dot{Y}_r(0) \mid D = 1 \right]. \quad (2.8)$$

Estimating the first term in (2.8) is easy because we observe $\dot{Y}_r(1)$ when $D = 1$. More precisely, define the same transformation in the observed variable Y_r :

$$\dot{Y}_r \equiv Y_r - \frac{1}{S-1} \sum_{q=1}^{S-1} Y_q \equiv Y_r - \bar{Y}_{pre} \quad (2.9)$$

When $D = 1$, $\dot{Y}_r = \dot{Y}_r(1)$, and so $E[\dot{Y}_r(1) \mid D = 1] = E(\dot{Y}_r \mid D = 1)$. The latter is trivially identified (as usual, under a suitable sampling scheme). Notice that, in the simple $T = 2$ case with $S = 2$, $\dot{Y}_2 = Y_2 - Y_1$, the difference from period one to period two.

The difficult term in identifying τ_r is $E[\dot{Y}_r(0) \mid D = 1]$. The unconditional parallel trends assumption implies that $E[\dot{Y}_r(0) \mid D = 1] = E[\dot{Y}_r(0) \mid D = 0]$. Here we follow the recent literature, for example CS (2021), and assume that parallel trends holds after conditioning on observed pre-treatment covariates.

Assumption CPTC (Conditional Parallel Trends, Common Timing): For observed covariates $\mathbf{X}$,

$$E[Y_t(0) - Y_1(0) \mid D, \mathbf{X}] = E[Y_t(0) - Y_1(0) \mid \mathbf{X}], \quad t = 2, \dots, T. \quad \square \quad (2.10)$$

Conceptually, we can think of the CPTC assumption as meaning that parallel trends holds within sub-partitions of the population as defined by $\mathbf{X}$. We do not require that the PT assumption hold across the entire population.

Simple algebra shows that (2.10) is the same as assuming $E[Y_t(0) - Y_s(0) \mid D, \mathbf{X}] =$

$E[Y_t(0) - Y_s(0) \mid \mathbf{X}]$ for all $t \neq s$. Wooldridge (2025b) used very similar assumptions, along with linearity of conditional means, to derive identification of the τ_r . Here, we are interested in applying methods other than linear regression adjustment to the transformed outcomes in (2.9). Assumption CPTC allows us to identify $E[\dot{Y}_r(0) \mid D = 1]$. To see how, first note that, by iterated expectations,

$$E[\dot{Y}_r(0) \mid D = 1] = E\left\{E[\dot{Y}_r(0) \mid D = 1, \mathbf{X}] \mid D = 1\right\} \quad (2.11)$$

Next, write

$$\dot{Y}_r(0) = (S - 1)^{-1} \sum_{q=1}^{S-1} [Y_r(0) - Y_q(0)].$$

Then, by CPTC,

$$\begin{aligned} E[\dot{Y}_r(0) \mid D = 1, \mathbf{X}] &= (S - 1)^{-1} \sum_{q=1}^{S-1} E[Y_r(0) - Y_q(0) \mid D = 1, \mathbf{X}] \\ &= (S - 1)^{-1} \sum_{q=1}^{S-1} E[Y_r(0) - Y_q(0) \mid D = 0, \mathbf{X}] \\ &= E\left[Y_r(0) - (S - 1)^{-1} \sum_{q=1}^{S-1} Y_q(0) \mid D = 0, \mathbf{X}\right] \\ &= E[\dot{Y}_r(0) \mid D = 0, \mathbf{X}]. \end{aligned} \quad (2.12)$$

The conclusion in equation (2.12) is simple but important. It says that, in terms of the potential outcome $\dot{Y}_r(0)$, treatment D is unconfounded conditional on $\mathbf{X}$. Assumption NAC ensures that the period-specific ATTs can be expressed as in (2.8). This means that, for a post-intervention period r , we have turned the difference-in-differences problem into a standard problem of estimating an ATT in a cross-sectional population.

Using the fact that $Y_q = Y_q(0)$ when $D = 0$, (2.12) implies that,

$$E[\dot{Y}_r(0) \mid D = 1, \mathbf{X}] = E(\dot{Y}_r \mid D = 0, \mathbf{X})$$

Now the argument is the same as in a standard cross section treatment effects setting:

By iterated expectations,

$$\begin{aligned} E[\dot{Y}_r(0) \mid D = 1] &= E[E(\dot{Y}_r \mid D = 0, \mathbf{X}) \mid D = 1] \\ &\equiv E[\dot{m}_{0r}(\mathbf{X}) \mid D = 1], \end{aligned} \quad (2.13)$$

where $\dot{m}_{0r}(\mathbf{X}) \equiv E(\dot{Y}_r \mid D = 0, \mathbf{X} = \mathbf{x})$ is the conditional mean of the observed variable $\dot{Y}_r$ for the control group. This function is nonparametrically identified on $\text{Supp}(\mathbf{X} \mid D = 0)$, the support of the covariates for the control group. To ensure that $E[\dot{m}_{0r}(\mathbf{X}) \mid D = 1]$ can be computed without extrapolating to covariate values outside $\text{Supp}(\mathbf{X} \mid D = 0)$, we impose a standard overlap assumption.

Assumption OVLC (Overlap, Common Timing): Define the propensity score

$$p(\mathbf{x}) = P(D = 1 \mid \mathbf{X} = \mathbf{x}), \mathbf{x} \in \text{Supp}(\mathbf{X}). \quad (2.14)$$

Then

$$p(\mathbf{x}) < 1, \mathbf{x} \in \text{Supp}(\mathbf{X}). \quad \square \quad (2.15)$$

The previous derivations and discussion prove the following.

Theorem 2.1 *Under Assumption NAC, τ_r can be expressed as in (2.8) for $r = S, \dots, T$. Under Assumption CPTC, D is unconfounded (in the conditional mean sense) with respect to $\dot{Y}_r(0)$ conditional on $\mathbf{X}$. When we add Assumption OVLC, the parameters τ_r , $r = S, \dots, T$, are identified. $\square$*

3 Estimation in the Common Timing Case

Given the identification result stated in Theorem 2.1, estimation of the τ_r is straightforward. Once the outcome is transformed as in (2.9), we can apply any estimation method. Essentially, this is the conclusion reached in Sant'Anna and Zhao (2020) in the $T = 2$ case. Earlier, Abadie (2005) proposed inverse probability weighting when $T = 2$.

Assume we observe a random sample of size N from the cross section. The observed outcome can be expressed as

$$Y_{it} = (1 - D_i) \cdot Y_{it}(0) + D_i \cdot Y_{it}(1), \quad (3.1)$$

where the i subscript denotes unit i . For each i , we observe $\{(Y_{it}, D_i, \mathbf{X}_i) : i = 1, 2, \dots, N\}$. To exploit the unconfoundedness result in Theorem 2.1, we simply need to obtain the transformed data. For each unit i , define

$$\dot{Y}_{ir} = Y_{ir} - \frac{1}{(S-1)} \sum_{q=1}^{S-1} Y_{iq} \equiv Y_{ir} - \bar{Y}_{i,pre} \quad (3.2)$$

Then, for any $r \in \{S, \dots, T\}$, we can apply any standard treatment effect (TE) estimator to the data

$$\{(\dot{Y}_{ir}, D_i, \mathbf{X}_i) : i = 1, 2, \dots, N\}.$$

A common TE estimator is (linear) “regression adjustment” (RA), which means estimating separate regression functions for the control and treated units. While the RA estimator of τ_r can be characterized using imputation, as shown in Wooldridge (2025b), it can instead be obtained from a single regression using all observations in period r :

$$\dot{Y}_{ir} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot (\mathbf{X}_i - \bar{\mathbf{X}}_1), i = 1, 2, \dots, N, \quad (3.3)$$

where $\hat{\tau}_r$ is the coefficient on D_i and $\bar{\mathbf{X}}_1 = N_1^{-1} \sum_{i=1}^N D_i \mathbf{X}_i$ is the vector of averages over the treated units. Obtaining $\hat{\tau}_r$ from single regression means that one can use standard software once the $\dot{Y}_{ir}$ have been obtained. This leads to simple inference for $\hat{\tau}_r$, allowing easy computation of standard errors robust to any kind of heteroskedasticity. Also, it is often easy to account for the sampling variation in $\bar{\mathbf{X}}_1$ as an estimator of $\mu_1 \equiv E(\mathbf{X} \mid D = 1)$. Issues of clustering standard errors are relatively easy to deal with given we have a standard cross-sectional regression.

Because of the representation of $\dot{Y}_{ir}$ in (3.2), there is a simple characterization of $\hat{\tau}_r$. All of the coefficients in (3.3) are obtained by differencing the coefficients from two separate regressions. In the first, Y_{ir} is regressed on all variables in (3.3). Then $\bar{Y}_{i,pre}$ is regressed on the same set of variables, and these coefficients are subtracted from the first. In particular, $\hat{\tau}_r$ is obtained as the difference between two standard ATT estimators using regression adjustment. The first uses observations in period r only, and the second uses the average of Y_{iq} over the pre-treatment periods. Without covariates, the estimator would be

$$\hat{\tau}_r = (\bar{Y}_{1r} - \bar{Y}_{0r}) - (\bar{Y}_{1,pre} - \bar{Y}_{0,pre}), \quad (3.4)$$

where the first subscript indicates treated or control status. This has a clear interpretation as a DiD estimator.

Recognizing that an estimator can be obtained from (3.3) has additional benefits. For example, if D_i is independent of $\dot{Y}_{ir}(0)$ —so that the unconditional PT assumption holds—then the covariates $\mathbf{X}_i$ need not be included in (3.3) for the coefficient on D_i to consistently estimate τ_r . If, in addition, D_i is independent of $\mathbf{X}_i$, the regression in (3.3) still can be used to improve efficiency over the simple estimator without the covariates. As discussed in Negi and Wooldridge (2021), such improvements are possible if $\mathbf{X}_i$ helps predict $\dot{Y}_{ir}$. In many cases, $\mathbf{X}_i$ may not have much predictive power for $\dot{Y}_{ir}$ even though it might predict the level, Y_{ir} , well.

It turns out there is another useful algebraic equivalence. Suppose we act *as if* the following conditional expectation holds for all population units and time periods:

$$\begin{aligned} E(Y_t | D, \mathbf{X}) = & \alpha + \mathbf{X}\beta + \gamma D + (D \cdot \mathbf{X})\delta + \sum_{r=2}^T \theta_r f r_t + \sum_{r=2}^T (f r_t \cdot \mathbf{X}) \pi_r \\ & + \sum_{r=S}^T \tau_r (D \cdot f r_t) + \sum_{r=S}^T (D \cdot f r_t) (\mathbf{X} - \mu_1) \eta_r, \quad t = 1, \dots, T, \end{aligned} \quad (3.5)$$

where $f r_t$ is a time period dummy equal to one if $r = t$ and zero otherwise. The interaction

$D \cdot fr_t$ is the treatment indicator for time period r . Equation (3.3) suggests a pooled OLS regression across all i and t :

$$Y_{it} \text{ on } 1, \mathbf{X}_i, D_i, D_i \cdot \mathbf{X}_i, f2_t, \dots, fT_t, f2_t \cdot \mathbf{X}_i, \dots, fT_t \cdot \mathbf{X}_i \\ D_i \cdot fS_t, \dots, D_i \cdot fT_t, D_i \cdot fS_t \cdot (\mathbf{X}_i - \bar{\mathbf{X}}_1), \dots, D_i \cdot fT_t \cdot (\mathbf{X}_i - \bar{\mathbf{X}}_1) \quad (3.6)$$

The estimated treatment effects, say $\tilde{\tau}_r$, are the coefficients on the treatment dummies $D_i \cdot fS_t, \dots, D_i \cdot fT_t$. Wooldridge (2025b) shows that the $\tilde{\tau}_r$ are numerically identical to a two-stage imputation approach based on the levels, Y_{it} . It turns out that the $\tilde{\tau}_r$ are also equivalent to the $\hat{\tau}_r$ obtained by using the transformed outcome variable, $\dot{Y}_{ir}$, one period at a time.

Theorem 3.1 *Let $\hat{\tau}_r$, $r = S, S + 1, \dots, T$ be the coefficients on D_i in the separate regressions (3.3) and let $\tilde{\tau}_r$ be the coefficients on $D_i \cdot fS_t, \dots, D_i \cdot fT_t$ from (3.6). Then $\tilde{\tau}_r = \hat{\tau}_r$, $r = S, S + 1, \dots, T$. Moreover, the coefficient vector on $D_i \cdot fr_t \cdot (\mathbf{X}_i - \bar{\mathbf{X}}_1)$ in (3.6) is identical to that on $D_i \cdot (\mathbf{X}_i - \bar{\mathbf{X}}_1)$ in (3.3). $\square$*

The proof of Theorem 3.1 can be found in Appendix A. The equivalence is valuable for a couple of reasons. First, it shows that two different ways to approach identification under the same set of assumptions—that in Wooldridge (2025b) and the approach we use here—lead to the same estimation methods. Second, Wooldridge (2025b, Theorem 6.2) shows that, under standard assumptions on the implied error term, which includes a unit-specific unobserved effect and a time-varying component, the estimators from (3.6) are both best linear unbiased and asymptotically efficient with T fixed and $N \rightarrow \infty$, under random sampling across i . This establishes that the transformation used in (3.3) does not discard useful information.

Given the equivalence of our transformation approach and the OLS estimator pooled across i and t , what use is the former? Importantly, it allows us to use other treatment

effects estimators beyond regression adjustment. For example, we can apply IPW or IPWRA using the cross-sectional data $\{(\dot{Y}_{ir}, D_i, \mathbf{X}_i) : i = 1, 2, \dots, N\}$. We can also apply propensity score matching or covariate matching.

Procedure 3.1 (Rolling Methods, Common Timing):

Step 1. For a given time period $r \in \{S, \dots, T\}$ and each unit i , compute $\dot{Y}_{ir}$ as in (3.2).

Step 2. Using all of the units, apply standard TE methods—such as linear RA, IPW, IPWRA, matching—to the cross section

$$\left\{ \left(\dot{Y}_{ir}, D_i, \mathbf{X}_i \right) : i = 1, \dots, N \right\}. \quad \square$$

Inference on a single cohort-time treatment effect is straightforward in Step 2 using built-in Stata commands such as `teffects`. The method of obtaining the IPWRA estimator using a weighted least squares regression, with IPW as a special case, is described in Section 19.4 of Wooldridge (2025a). After the rolling transformation, the IPWRA estimator can be viewed as a two-step M-estimator. Under standard regularity conditions for two-step M-estimation (e.g., Wooldridge, 2007, Newey and McFadden, 1994)—including overlap, finite second moments of the relevant estimating equations (e.g., $E[Y^2] < \infty$ and $E[\|X\|^2] < \infty$), nonsingularity of the associated moment matrices, and sufficient regularity of the first-step estimators—consistency and asymptotic normality follow. For IPWRA, consistency requires correct specification of either the propensity score model or the outcome model.

We can compare the transformation in equation (3.2) to that in CS (2021). In the common timing case, the CS transformation, for $r \geq S$, is

$$\dot{Y}_{ir} = Y_{ir} - Y_{i,S-1}, \tag{3.7}$$

so that $\dot{Y}_{ir}$ is a “long” difference. When $T=2$, the transformation is $\dot{Y}_2 = \dot{Y}_2 = \Delta Y_2 \equiv Y_2 - Y_1$. Thus, our approach encompasses and extends Abadie (2005) and Sant’Anna and Zhao (2020) in the panel data case.

4 Staggered Interventions

4.1 Some Units Never Treated

We now turn to the staggered intervention case. As in [Athey and Imbens \(2022\)](#) and [Wooldridge \(2023, 2025b\)](#), the potential outcomes are denoted

$$Y_t(g), g \in \{S, \dots, T, \infty\}, t \in \{1, 2, \dots, T\}, \quad (4.1)$$

where g indicates the first treatment period—thereby defining the treatment cohorts—and t is calendar time. The case $g = \infty$ indicates the potential outcome in the never-treated state. In other words, $Y_t(\infty)$ is the potential outcome at time t when a unit is not subjected to the intervention over the observed stretch of time. Listing potential outcomes that vary only by cohort and calendar time reflects the assumption of no reversibility with staggered entry.

We denote the group or cohort indicators by $\{D_S, D_{S+1}, \dots, D_T, D_\infty\}$, where $D_\infty = 1$ indicates the never-treated. These dummy variables are mutually exclusive and exhaustive: $D_S + \dots + D_T + D_\infty = 1$.

The ATTs of interest are now written as

$$\tau_{gr} = E[Y_r(g) - Y_r(\infty) | D_g = 1], r = g, \dots, T; g = S, \dots, T. \quad (4.2)$$

For each treated cohort g , τ_{gr} denotes the ATT in each subsequent period r .

To identify the τ_{gr} , we extend the trick for the common timing case by writing

$$\begin{aligned} Y_t(g) - Y_t(\infty) &= \left[Y_t(g) - \frac{1}{(g-1)} \sum_{s=1}^{g-1} Y_s(g) \right] - \left[Y_t(\infty) - \frac{1}{(g-1)} \sum_{s=1}^{g-1} Y_s(\infty) \right] \\ &\quad + \frac{1}{(g-1)} \sum_{s=1}^{g-1} [Y_s(g) - Y_s(\infty)] \end{aligned} \quad (4.3)$$

As in the common timing case, we make a no anticipation assumption so that the third

term can be dropped, which effectively allows us to use all available control units in each treated period. Here, we condition on the covariates so that not-yet-treated units can be used as part of the control group.

Assumption CNAS (Conditional No Anticipation, Staggered): For $g \in \{S, \dots, T\}$, $t \in \{1, \dots, g-1\}$ and covariates $\mathbf{X}$,

$$E[Y_t(g) | D_g = 1, \mathbf{X}] = E[Y_t(\infty) | D_g = 1, \mathbf{X}]. \quad \square \quad (4.4)$$

As in the common timing case, this assumption means that the “treatment” effects prior to the intervention are all zero. Because $s < g$ in the third sum, it follows that the expectation of the last term conditional on $D_g = 1$ is zero. Therefore,

$$\tau_{gr} = E[\dot{Y}_{rg}(g) | D_g = 1] - E[\dot{Y}_{rg}(\infty) | D_g = 1], \quad (4.5)$$

where $\dot{Y}_{rg}(g)$ and $\dot{Y}_{rg}(\infty)$ are defined as the first and second terms in (4.3), respectively. Note that the first subscript on $\dot{Y}_{rg}(g)$ and $\dot{Y}_{rg}(\infty)$ means that we are averaging all periods just prior to g and subtracting from the outcome in the current calendar time period r .

As before, the first term in equation (4.3) is easily estimated because we observe $\dot{Y}_{rg}(g)$ when $D_g = 1$. A parallel trends assumption, stated in terms of the never-treated state, is sufficient to identify $E[\dot{Y}_{rg}(\infty) | D_g = 1]$. We state the assumption conditional on a set of covariates, $\mathbf{X}$, with no covariates as a special case.

Assumption CPTS (Conditional PT, Staggered): For $\mathbf{D} = (D_S, \dots, D_T)$ and $t = 1, 2, \dots, T$,

$$E[Y_t(\infty) - Y_1(\infty) | \mathbf{D}, \mathbf{X}] = E[Y_t(\infty) - Y_1(\infty) | \mathbf{X}], \quad t = 2, \dots, T. \quad \square \quad (4.6)$$

This assumption is used in [Wooldridge \(2025b\)](#). Again, it is unconfoundedness of the treatment level, as given by $\mathbf{D}$, with respect to the trend in the untreated state, $Y_t(\infty) - Y_1(\infty)$. [Wooldridge \(2025b\)](#) used this assumption, along with linearity of conditional means,

to derive an imputation estimator and showed it was the same as a pooled OLS. Here, we show how it can be used to identify the τ_{gr} very generally.

With $\dot{Y}_{rg}(\infty)$ defined above,

$$\begin{aligned} E[\dot{Y}_{rg}(\infty) \mid D_g = 1, \mathbf{X}] &= \frac{1}{(g-1)} \sum_{s=1}^{g-1} E[Y_r(\infty) - Y_s(\infty) \mid D_g = 1, \mathbf{X}] \\ &= \frac{1}{(g-1)} \sum_{s=1}^{g-1} E[Y_r(\infty) - Y_s(\infty) \mid D_\infty = 1, \mathbf{X}] \quad (4.7) \\ &= E[\dot{Y}_{rg}(\infty) \mid D_\infty = 1, \mathbf{X}] \end{aligned}$$

where the second equality follows from CPTS and the third follows by taking the expectation outside the summation. We have shown the following.

Theorem 4.1 *Under Assumption CNAS, equation (4.5) holds. If we add Assumption CPTS, the cohort assignments, $\mathbf{D} = (D_S, \dots, D_T)$ are unconfounded with respect to $\dot{Y}_{rg}(\infty)$ (in the conditional mean sense), $g \in \{S, \dots, T\}$, $r \in \{g, \dots, T\}$, conditional on $\mathbf{X}$. $\square$*

Because the vector of cohort indicators is unconfounded with respect to $\dot{Y}_{rg}(\infty)$, Theorem 4.1 implies

$$E[\dot{Y}_{rg}(\infty) \mid D_\infty = 1, \mathbf{X}] = E[\dot{Y}_{rg}(\infty) \mid D_h = 1, \mathbf{X}], \quad h = S, \dots, T \quad (4.8)$$

We can combine this implication of CPTS with Assumption CNAS because, at time r , cohorts $h = r+1, \dots, T$ have yet to be treated. Therefore,

$$E[\dot{Y}_{rg}(\infty) \mid D_h = 1, \mathbf{X}] = E[\dot{Y}_{rg}(h) \mid D_h = 1, \mathbf{X}], \quad h = r+1, \dots, T \quad (4.9)$$

Combined, (4.8) and (4.9) mean that, in addition to the never-treated (NT) group, we can use treatment cohorts $h \in \{r+1, \dots, T\}$ in estimating $E[\dot{Y}_{rg}(\infty) \mid D_g = 1, \mathbf{X}]$. Incidentally, this derivation shows that if we only use the NT group as the control for each (g, r) pair, then we can drop the conditioning on $\mathbf{X}$ in Assumption CNAS. Later, we discuss what can be identified in period T without an NT group under CNAS.

We have established the following. For estimating $E[\dot{Y}_{rg}(\infty) \mid D_g = 1, \mathbf{X}]$ for $r \in \{g, \dots, T\}$, we can use cohorts $\{r+1, \dots, T, \infty\}$ as the control group. Define the indicator for the control group as $A_{r+1} \equiv D_{r+1} + D_{r+2} + \dots + D_T + D_\infty$. Then, within the subpopulation $D_g + A_{r+1} = 1$, D_g is unconfounded with respect to $\dot{Y}_{rg}(\infty)$, conditional on $\mathbf{X}$. Therefore, we can apply standard treatment effect estimators after transforming the observed outcome and conditioning on this subpopulation.

Naturally, we will need an overlap assumption in order to ensure identification when using methods that condition on covariates. For τ_{gr} and using all legitimate control groups under CNAS and CPTS, the overlap assumption is

Assumption OVLS (Overlap, Staggered Case): For cohorts $g \in \{S, \dots, T\}$ and time periods $r \in \{g, \dots, T\}$,

$$P(D_g = 1 \mid D_g + A_{r+1} = 1, \mathbf{X} = \mathbf{x}) < 1 \text{ for all } \mathbf{x} \in \text{Supp}(\mathbf{X}). \quad \square \quad (4.10)$$

This condition ensures that, within the subpopulation of cohort g plus the never-treated and not-yet-treated units at time r , every treated unit has a comparable control unit.

Given data on units again indexed by i , the following simple steps lead to a general analysis. Assumptions CNAS, CPTS, and the OVLS are in force.

Procedure 4.1 (Rolling Methods, Staggered Interventions):

Step 1. For a given $g \in \{S, \dots, T\}$ and time period $r \in \{g, \dots, T\}$, compute

$$\dot{Y}_{irg} \equiv Y_{ir} - \frac{1}{(g-1)} \sum_{s=1}^{g-1} Y_{is} \equiv Y_{ir} - \bar{Y}_{i,pre(g)} \quad (4.11)$$

Step 2. Choose as the control group the units with $A_{i,r+1} = 1$ (or, if desired, a subset, such as the NT group).

Step 3. Using the subset of data with $D_{ig} + A_{i,r+1} = 1$, apply standard TE methods – such as linear RA, IPW, IPWRA, matching – to the cross section

$$\left\{ \left(\dot{Y}_{irg}, D_{ig}, \mathbf{X}_i \right) : i = 1, \dots, N \right\},$$

with D_{ig} acting as the treatment indicator. $\square$

When $\mathbf{X}_i$ has high dimension, Procedure 4.1 implies that machine learning methods can be applied after obtaining the transformation in (4.11). See, for example, [Belloni et al. \(2014\)](#) and [Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey and Robins \(2018\)](#).

Interestingly, when $r = g$, so that τ_{gg} is the instantaneous effect of the intervention for treatment cohort g , applying linear RA to

$$\left\{ (\dot{Y}_{igg}, D_{ig}, \mathbf{X}_i) : i = 1, \dots, N \right\},$$

with all possible control units, reproduces the POLS estimator proposed by [Wooldridge \(2025b\)](#). When $r > g$ this is not the case, which means, under standard assumptions, the rolling approach we propose is inefficient for the dynamic effects. The trade-off is that we are able to apply many different kinds of estimators, including doubly robust and matching estimators.

Applying matching estimators in a rolling setting has distinct advantages over the practice of pre-matching all treated units with the never-treated group and then using standard DiD methods. For one, matches can be obtained from all not-yet-treated units in a given

time period, possibly resulting in better matches. Also, if we apply standard matching software, the standard errors will reflect the uncertainty generated due to matching.

4.2 Comparison with Alternative Methods

An approach that has a similar flavor to Procedure 4.1 is the “build-the-trend” (BTT) estimator in [Callaway \(2023\)](#), which uses all pre-treatment periods to weight pairwise differences and uses all not-yet-treated units as controls. However, the BTT estimator is a formula based on inverse probability weighting; it is not motivated as just one of many special cases that is justified by the conclusion of unconfounded treatment. The estimator is different from that in Procedure 4.1 when we use IPW; moreover, BTT does not handle heterogeneous trends, an extension of our approach that we consider in Section 5.

Procedure 4.1 can be compared with the [Callaway and Sant’Anna \(2021\)](#) approach with staggered interventions. CS (2021) suggest using a long difference of the form

$$\mathring{Y}_{irg} \equiv Y_{ir} - Y_{i,g-1} \tag{4.12}$$

and then choosing control groups from cohorts $\{r + 1, \dots, T, \infty\}$. The transformation in (4.12) ignores control periods prior to $g - 1$, whereas (4.11) gives weight $\frac{1}{(g-1)}$ to all pre-treatment outcomes.

A sketch of an identification argument that mimics that in Section 4.1 for the CS (2021) approach is instructive. For $g \leq t \leq T$, write

$$Y_t(g) - Y_t(\infty) = [Y_t(g) - Y_{g-1}(g)] - [Y_t(\infty) - Y_{g-1}(\infty)] + [Y_{g-1}(g) - Y_{g-1}(\infty)].$$

Imposing a no anticipation assumption

$$E[Y_{g-1}(g) - Y_{g-1}(\infty) \mid D_g = 1] = 0,$$

implies

$$\begin{aligned}\tau_{gt} &= E[Y_t(g) - Y_t(\infty) \mid D_g = 1] \\ &= E[Y_t(g) - Y_{g-1}(g) \mid D_g = 1] - E[Y_t(\infty) - Y_{g-1}(\infty) \mid D_g = 1].\end{aligned}\tag{4.13}$$

The first part is always identified because

$$E[Y_t(g) - Y_{g-1}(g) \mid D_g = 1] = E[Y_t - Y_{g-1} \mid D_g = 1],$$

and so we can average the “long” difference of the units in cohort g . For the second part of (4.13), we can impose a conditional parallel trends assumption

$$E[Y_t(\infty) - Y_{g-1}(\infty) \mid D_S, \dots, D_T, X] = E[Y_t(\infty) - Y_{g-1}(\infty) \mid X],\tag{4.14}$$

which is the same as saying that the vector of cohort assignments, $(D_S, \dots, D_T)$, is unconfounded with respect to $Y_t(\infty) - Y_{g-1}(\infty)$ conditional on X .

Now the argument is the same as in Section 4.1: we can apply any treatment effect estimator based on unconfoundedness, and we can use the not-yet-treated units at time t —so cohorts $h \in \{t + 1, t + 2, \dots, T, \infty\}$ as the control group.

As discussed in [Callaway and Sant’Anna \(2021\)](#), equation (4.14) only requires the CPT assumption to hold starting from period $g - 1$ and allows violations before $g - 1$. While [Marcus and Sant’Anna \(2021\)](#) describe a potential bias–variance trade-off from using more pre-treatment periods, such a trade-off does not always exist. [Wooldridge \(2025b\)](#) notes that if CPT fails into the treated periods, the CS (2021) estimator can have more bias than the extended TWFE (ETWFE) estimator. Like the ETWFE estimator, our approach uses all pre-treatment periods, and so it can have less bias than CS (2021) when CPT fails. Unfortunately, we cannot know when that will happen.

Information Set Comparison. Assuming the parallel trends assumption holds, we next compare the information sets used by the rolling, CS, and Wooldridge approaches. Table 1 summarizes the pre-treatment information used by each estimator. To illustrate the comparison, we consider treatment cohorts $g \in \{4, 5, 6\}$ and focus on estimating $ATT(4, 5)$, denoted by $\star$. Shaded cells indicate the pre-treatment outcome observations used by each estimator when estimating the target parameter.

Table 1: Pre-Treatment Period Information Used by Each Method

time	CS (2021)			Rolling Method			Wooldridge (2025b)		
	$g = 4$	$g = 5$	$g = 6$	$g = 4$	$g = 5$	$g = 6$	$g = 4$	$g = 5$	$g = 6$
1									
2									
3									
4									
5	$\star$			$\star$			$\star$		
6									

The regression-based approach in Wooldridge (2025b) uses all available pre-treatment information and can be the most efficient among the three under a correctly specified linear conditional mean and standard error-components assumptions. However, this efficiency comes at the cost of potential bias if the conditional mean is misspecified.

The rolling method uses less information than the Wooldridge approach because it rules out *bad comparisons*; for example, when estimating $ATT(4, 5)$, both the rolling and CS approaches exclude group 5 at $t = 5$, since it is already treated. This may sacrifice some precision relative to Wooldridge, but, like CS, it accommodates doubly robust estimation. At the same time, compared with CS, the rolling method uses more pre-treatment information through the demeaning transformation, which can improve efficiency.

In this respect, our rolling estimator can be viewed as a middle ground between the CS (2021) and Wooldridge (2025b) approaches: it preserves the clean-comparison logic of CS while using more pre-treatment information, and it sacrifices some of the information used by Wooldridge in exchange for greater robustness to model misspecification.

4.3 All Units Eventually Treated

As in Wooldridge (2023, 2025b), we can handle situations where all units are treated by $t = T$ by modifying the parallel trends assumptions and changing the specifics of the estimation. Rather than stating the CPT assumption in terms of the NT state, $Y_t(\infty)$, it is initially stated in terms of $Y_t(T)$, the state of not being treated until the final time period. Now all of the treatment effects are, initially, defined relative to $Y_t(T)$: $E[Y_r(g) - Y_r(T) | D_g = 1]$, $g \in \{S, \dots, T-1\}$, $r \in \{g, \dots, T\}$. We can no longer estimate a treatment effect for the final treated cohort because there are no available control units.

As discussed in Wooldridge (2025b), by no anticipation it follows that for $r < T$, $E[Y_r(g) - Y_r(T) | D_g = 1] = E[Y_r(g) - Y_r(\infty) | D_g = 1]$ and so, except for the final time period, the ATTs are interpreted as when we have a never-treated group. Unlike other approaches, in the final time period we can estimate the effect of being treated in earlier periods relative to being treated in the last period. This is possible using our transformation or the CS transformation, but CS (2021) does not explore this possibility.

In terms of estimation, the modifications to Procedure 4.1 are straightforward. In particular, $\dot{Y}_{irg}$ is computed as in (4.11) but only for $g \in \{S, \dots, T-1\}$. In steps (2) and (3), we drop $D_{i\infty}$ everywhere – which means still choosing as the control group for cohort g in period r those units not yet treated. Then, for each $g \in \{S, \dots, T-1\}$ and for each period $r \in \{g, \dots, T\}$, we apply standard treatment effect estimators, as before. When $r = T$, $D_T = 1$ acts as the only control group for all cohorts first treated prior to period T .

4.4 Violations of No Anticipation. Unbalanced Panels

The no anticipation assumption requires that, prior to the first intervention period for a given treatment cohort, the potential outcomes are the same (on average) as in the never-treated state. This assumption can fail if units know that a program or policy change is approaching prior to its being actually implemented. If the NA assumption is in doubt,

one can drop one or more periods prior to the intervention time and redo the analysis as a robustness check.

As an example, suppose a cohort is first treated in $g = 5$. In Procedure 4.1, one would average over periods $\{1, 2, 3, 4\}$ in obtaining the average to remove for Y_{i5} (or, one would remove a unit-specific linear trend, as in Procedure 5.1). Instead, one might drop period four altogether, or maybe even periods three and four. Any precise recommendation is context specific. It is very easy to apply any of the procedures we have recommended to cases where time periods are skipped.

Another issue that often arises in practice is unbalanced panels. With time-constant controls, unbalancedness would typically arise because of missing data on Y_{it} , possibly due to attrition. If data are missing on Y_{it} for some time periods for unit i , the demeaning or detrending is simply applied to the observed data. The mechanics of the procedure are then exactly the same. For treatment cohort g in period r , the transformed outcome ($\dot{Y}_{irg}$ or $\ddot{Y}_{irg}$) can only be used if there are enough observed data in the periods $t < g$ to compute an average (one period) or a linear trend (two periods). Of course, Y_{ir} must also be observed.

It is natural to wonder when ignoring the reason the panel is unbalanced does not cause systematic bias. Because our method removes unit-specific averages in Procedure 4.1, selection is allowed to depend on unobserved time-constant heterogeneity—just like with the usual fixed effects estimator. Selection cannot be systematically related to the shocks to $Y_{it}(\infty)$ —again, just as with the FE estimator.

5 Heterogeneous Trends

One way to test for violation of the PT assumption is to estimate placebo treatment effects prior to the intervention. [Callaway and Sant’Anna \(2021\)](#) take this approach using their differencing method. Here, we can apply Procedure 3.1 or 4.1 to pre-treatment periods and test for effects prior to the intervention. For cohort g , it makes sense to split pre-treatment

periods, $\{1, 2, \dots, g-1\}$, into roughly equal sizes. Then, the never-treated group, or any of the groups not yet treated in $\{1, \dots, g-1\}$ can be used as the controls. Under the null hypothesis of (conditional) PT, the tests should not find a “treatment” effect.

As motivation for adjusting Procedure 4.1 (with common timing being a special case) to allow for heterogeneous trends, express the conditional PT assumption as

$$E[Y_t(\infty)|\mathbf{D}, \mathbf{X}] = q_\infty(\mathbf{X}) + \sum_{g=S}^T D_g q_g(\mathbf{X}) + m_t(\mathbf{X}), t = 1, \dots, T, \quad (5.1)$$

where $q_g(\cdot)$ and $m_t(\cdot)$ are functions of the covariates, with the first not changing across time and the second not depending on the treatment cohort. As a normalization, $m_1(\mathbf{x}) \equiv 0$ for all $\mathbf{x} \in \text{Supp}(\mathbf{X})$. It is easily seen that

$$E[Y_t(\infty) - Y_1(\infty)|\mathbf{D}, \mathbf{X}] = m_t(\mathbf{X}), t = 2, \dots, T,$$

which does not depend on $\mathbf{D}$. Moreover, for $r \geq g$, the definition of $\dot{Y}_{rg}(\infty)$ implies that

$$E[\dot{Y}_{rg}(\infty) | D_\infty = 1, \mathbf{X}] = m_r(\mathbf{X}) - \frac{1}{(g-1)} \sum_{s=1}^{g-1} m_s(\mathbf{X}) \equiv \dot{m}_{rg}(\mathbf{X}) \quad (5.2)$$

and so $\dot{m}_{rg}(\cdot)$ is the conditional mean function implicit in the methods from Section 4 that use a conditional mean specification (RA or IPWRA). Because $\dot{m}_{rg}(\cdot)$ can take on positive and negative values, regression-based methods essentially assume that $\dot{m}_{rg}(\cdot)$ can be approximated by a function linear in parameters.

The representation in (5.1) suggests a way to relax parallel trends for cohorts where we have at least two pre-treatment time periods. We replace (5.1) with the following assumption.

Assumption CHT (Conditional Heterogeneous Trends): For $\mathbf{D} = (D_S, \dots, D_T)$ and $t = 1, 2, \dots, T$,

$$E[Y_t(\infty)|\mathbf{D}, \mathbf{X}] = \eta_S(D_S \cdot t) + \dots + \eta_T(D_T \cdot t) + q_\infty(\mathbf{X}) + \sum_{g=S}^T D_g q_g(\mathbf{X}) + m_t(\mathbf{X}). \quad \square \quad (5.3)$$

Assumption CHT allows a separate linear trend in the never-treated state for each treatment cohort. It is easy to see that

$$E[Y_t(\infty) - Y_{t-1}(\infty) | \mathbf{D}, \mathbf{X}] = \eta_g D_g + \cdots + \eta_T D_T + [m_t(\mathbf{X}) - m_{t-1}(\mathbf{X})] \quad (5.4)$$

and so PT, even conditional on $\mathbf{X}$, fails. Because the trend in the never-treated state is systematically related to cohort, the estimation approaches in Sections 3 and 4 are no longer consistent.

Instead, we can linearly detrend the outcome, unit by unit, to remove the relationship between $Y_t(\infty)$ and cohort assignment. For any i , we can write

$$Y_{it}(\infty) = \mathbf{h}(\mathbf{D}_i, \mathbf{X}_i) + \mathbf{D}_i \cdot t \cdot \eta + m_t(\mathbf{X}_i) + U_{it}(\infty) \quad (5.5)$$

$$E[U_{it}(\infty) | \mathbf{D}_i, \mathbf{X}_i] = 0,$$

where $\mathbf{h}(\mathbf{D}_i, \mathbf{X}_i)$ does not vary across t . For any $t \geq 2$, we can eliminate both $\mathbf{h}(\mathbf{D}_i, \mathbf{X}_i)$ and $\mathbf{D}_i \cdot t \cdot \eta$ using unit-specific linear detrending.

Equation (5.5) is an example of a heterogeneous (or random) trend model of the kind discussed in Wooldridge (2010, Section 11.7.2). Appendix B provides further details on the procedure for removing unit-specific linear trends by regressing the outcome on a linear function of time over the pre-treatment periods. The residuals from this regression define the detrended outcome variable, denoted as $\ddot{Y}_{irg}(\infty)$. We then show that the treatment cohort indicators, D_{ig} , are unconfounded with respect to these detrended variables, conditional on $\mathbf{X}_i$. Moreover, by Assumption CNAS, cohorts not yet treated by period r —those with $h > r$ —can be used as the control group. The argument then follows the same logic as in Section 4: under Assumptions CNAS, CHT, and OVLS, we have justified the following procedure as yielding consistent estimates of τ_{gr} .

Procedure 5.1 (Staggered Entry):

Step 1. For a specified cohort $g \in \{S, \dots, T\}$, run the unit-specific regressions

$$Y_{it} \text{ on } 1, t, \quad t = 1, \dots, g - 1 \quad (5.6)$$

For $r \in \{g, \dots, T\}$, compute the out-of-sample predictions $\hat{Y}_{irg}$ and the residuals (detrended variables) $\ddot{Y}_{irg} \equiv Y_{ir} - \hat{Y}_{irg}$. (One need not do this for units treated prior to period g .)

Step 2. Same as Procedure 4.1.

Step 3. Identical to Procedure 4.1 with $\ddot{Y}_{irg}$ replacing $\dot{Y}_{irg}$. $\square$

Procedure 5.1 is very easy to implement, only requiring many unit-specific simple regressions on a constant and time trend. The common timing case is especially easy because the regression in (5.6) is done with $g = S$ only and then the detrended outcomes $\ddot{Y}_{ir}$ are used in standard treatment effect estimation for $r = S, \dots, T$.

In the simplest case where Procedure 5.1 can be applied, with $T = 3$ and intervention at $S = 3$ (without covariates), the resulting estimator of τ_3 is easily obtained after simple algebra:

$$[(\bar{Y}_{13} - \bar{Y}_{12}) - (\bar{Y}_{03} - \bar{Y}_{02})] - [(\bar{Y}_{12} - \bar{Y}_{11}) - (\bar{Y}_{02} - \bar{Y}_{01})], \quad (5.7)$$

where the first subscript on the average is one for treated and zero for control, and the second subscript indicates time period. The first term in brackets is the usual two-period DiD estimator if the first time period is ignored. The second term is an estimate of the difference in trends prior to the intervention – often interpreted as estimating a placebo effect. The estimator in (5.7) is an example of a difference-in-difference-in-differences estimator. Procedure 5.1 allows one to control for covariates when removing an estimated pre-intervention difference in trends is not sufficient to uncover a causal effect.

Removing unit-specific trends provides additional resiliency to unbalanced panels, as we are now allowing for two sources of heterogeneity—level and trend—to be correlated

with selection into the sample.

A final point worth mentioning is that the unit-specific linear regressions in (5.6) do not create an incidental parameters problem when the number of time periods is small. These regressions do not require us to obtain “good” estimates of the unit-specific trends; we only need to remove the unit-specific heterogeneity specified in Assumption CHT. The logic is the same as when we use all time periods to remove unit-specific trends, as in Wooldridge (2010, Section 11.7.2).

6 Application

In this section, we revisit the literature examining the effects of Walmart’s entry on local labor markets (Basker, 2005, Neumark et al., 2008, Brown and Butts, 2025). Prior studies highlight concerns about potential violations of the parallel trends assumption. For example, Brown and Butts (2025) provide visual evidence suggesting that much of the estimated effect could be due to pre-existing trends rather than the impact of Walmart’s entry (Rambachan and Roth, 2023). If Walmart’s entry is driven by county-level trends in economic fundamentals, observed differences in employment may reflect these trends rather than the causal effect of the store openings.

6.1 Data

As shown in Section 5, when violations of parallel trends are driven by unit-specific linear trends, our rolling method can produce stable estimates by explicitly removing these trends. We evaluate this feature using the dataset constructed by Brown and Butts (2025), who follow Basker (2005) in using County Business Patterns data from 1964 and 1977–1999.

Following Brown and Butts (2025), the sample is restricted to counties with total employment greater than 1,500 in 1964 and non-negative aggregate employment growth between 1964 and 1977. The Walmart store opening indicator is derived from the geographic dataset of openings in Arcidiacono, Ellickson, Mela and Singleton (2020). Counties whose

first Walmart opened before 1986 are excluded. This yields a balanced panel of 1,277 counties over 23 years, with each county observed for at least nine pre-treatment years and up to fourteen post-treatment years.

Table 2 reports descriptive statistics for the variables used in our analysis. Treatment cohorts are defined by the year of the first Walmart opening, and the outcome is county-level retail employment. The covariates are the 1980 population shares employed in manufacturing, above the poverty line, and with a high school diploma.

Table 2: Descriptive Statistics of Variables

Variable	Obs	Mean	Std.	Min	Max
log(Retail Employment)	29,371	7.76	1.28	4.58	12.92
Share of Population above Poverty Line	29,371	0.85	0.06	0.52	0.96
Share of Population in Manufacturing	29,371	0.10	0.05	0.01	0.29
Share of Population with a High School Diploma	29,371	0.09	0.03	0.02	0.19
Treated Cohort				1986	1999
Counties	1,277				

6.2 Estimation Results

For event time (relative period) $r = t - g$, define $WATT(r)$ as the cohort-size-weighted average of the identified cohort–time ATTs:

$$WATT(r) = \sum_{g \in \mathcal{G}_r} w(g, r) ATT(g, g + r),$$

where $\mathcal{G}_r$ is the set of cohorts for which $ATT(g, g + r)$ is identified, $w(g, r) = N_g / N_{\mathcal{G}_r}$, and $N_{\mathcal{G}_r} = \sum_{g \in \mathcal{G}_r} N_g$. The set of estimable event times depends on the outcome transformation: $r = -1$ is excluded under demeaning, while $r = -2$ and $r = -1$ are excluded under detrending, because fitting a unit-specific mean and a unit-specific linear trend requires at least one and two pre-treatment periods, respectively.

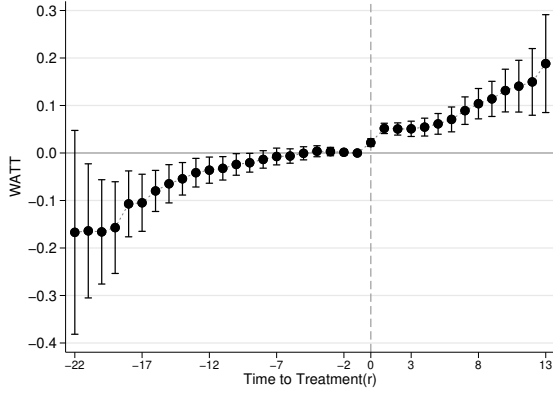
For inference on $WATT(r)$ and the event-study path $\{WATT(r) : r \in \mathcal{R}\}$, we rely on the influence-function representations implied by the stacked estimating equations. We then use these representations to implement a computationally efficient multiplier bootstrap

for standard errors and simultaneous confidence bands. The procedure is implemented in the user-written Stata command `lwdid`; see [Hur, Lee and Wooldridge \(2026\)](#) for details on the Stata implementation. The estimation and inference procedures are described in Appendices [D](#) and [E](#), respectively. In particular, Sections [E.2–E.4](#) derive the influence-function representations for the rolling RA, rolling IPWRA, and rolling IPW estimators.

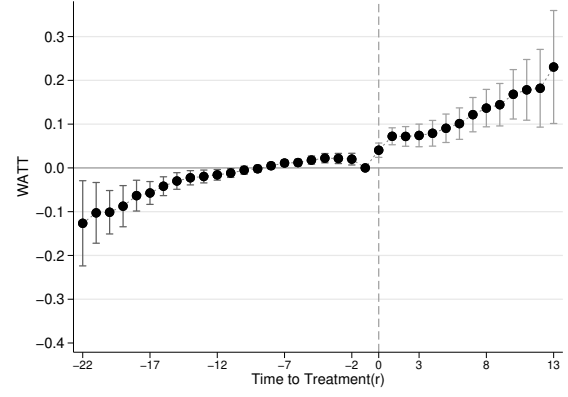
In [Figure 1](#), we report $WATT(r)$ estimates of the effect of Walmart entry on county-level retail employment in logs, obtained from the rolling IPWRA estimator with unit-specific demeaning and detrending, and from the CS ([2021](#)) doubly robust estimator. We also report 95% simultaneous confidence bands for the entire event-study path, constructed via influence-function-based multiplier bootstrap; see [Appendix E](#) for details.

Panels (a) and (b) of [Figure 1](#) use different outcome transformations and are therefore normalized differently. The CS ([2021](#)) estimates in Panel (a) are normalized to the last pre-treatment period, $r = -1$. In Panel (b), the rolling IPWRA estimates use outcomes demeaned by each county’s pre-treatment mean and therefore compare treated and comparison counties in deviations from their own pre-treatment averages. Both approaches rely on their respective parallel trends conditions for identification. However, the upward pre-treatment pattern in both panels indicates differential pre-treatment dynamics between treated and comparison counties, raising concerns about violations of the relevant parallel trends conditions and the interpretation of the post-entry estimates.

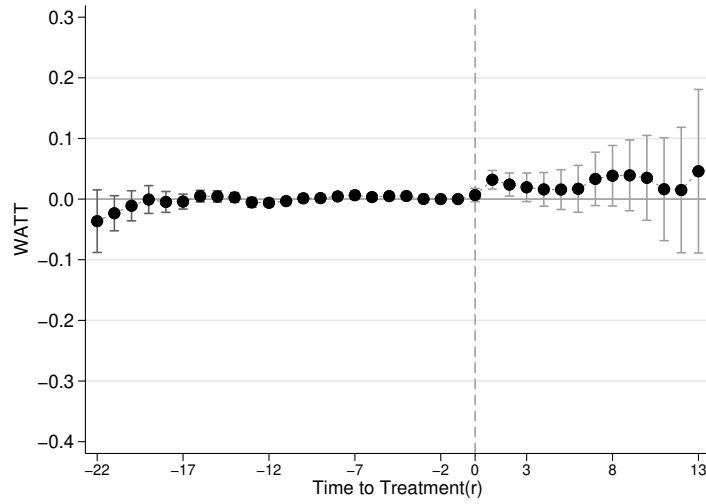
Panel (c) addresses this concern by using the rolling IPWRA estimator with detrending, which removes fitted county-specific linear trends based on pre-treatment observations. The resulting pre-treatment estimates are closer to zero and statistically insignificant, suggesting that the detrending procedure effectively mitigates heterogeneous pre-treatment trends across groups. The detrended specification therefore provides a more credible basis for interpreting the post-entry estimates as causal effects.



Panel (a) CS (2021) Approach



Panel (b) Rolling IPWRA with demeaning



Panel (c) Rolling IPWRA with detrending

Figure 1: Effects of Walmart Openings on $\log(\text{Retail Employment})$

The post-entry estimates from the detrended specification are positive in the short run but more modest than those from CS (2021) or the demeaned rolling estimator. For example, $\text{WATT}(1) = 0.032$ ($\text{SE} = 0.005$), implying a 3.2% increase in retail employment one year after entry, compared with approximately 5.2% under the CS estimator. This corresponds to about 210 additional retail jobs, which is well within the 150–300 workers typically employed by a Walmart store (Basker, 2005). At longer horizons, the detrended estimates return toward zero and are mostly statistically insignificant. This pattern is more

consistent with modest direct employment effects than with large spillover effects.

The full set of estimates is reported in Appendix D, and Appendix F.1 reports the corresponding results for wholesale employment. These findings highlight the empirical relevance of accounting for unit-specific trends when pre-treatment dynamics differ systematically across treated and comparison units. The results can be replicated using the Stata command `lwdid`.

7 Concluding Remarks

We propose a flexible transformation approach for treatment effect estimation in panel data settings, covering both common-timing and staggered interventions. A key advantage is that, once the outcome is transformed by removing unit-specific pre-treatment means, any standard treatment effect estimator—including regression adjustment, matching, doubly robust procedures, and causal machine learning methods—can be applied to the resulting sequence of cross-sectional samples, enabling consistent estimation of heterogeneous treatment effects across cohorts and exposure lengths. The transformation also extends naturally to unit-specific detrending, which accommodates heterogeneous linear trends and thereby relaxes the parallel trends assumption even after conditioning on covariates.

Our empirical application, which examines the effects of Walmart entry on local labor markets, illustrates how detrending can be useful in settings with unit-specific linear trends, where widely used conditional parallel-trends methods, such as CS (2021), are not designed to adjust for such patterns. Together with the accompanying Stata command `lwdid`, which implements the full procedure and multiplier-bootstrap inference, these features make the proposed rolling method a valuable tool for applied researchers.

References

- Abadie, A. (2005), “Semiparametric Difference-in-Differences Estimators,” *Review of Economic Studies*, 72, 1–19.
- Arcidiacono, P., Ellickson, P. B., Mela, C. F. and Singleton, J. D. (2020), “The Competitive Effects of Entry: Evidence from Supercenter Expansion,” *American Economic Journal: Applied Economics*, 12, 175–206.
- Athey, S. and Imbens, G. W. (2022), “Design-Based Analysis in Difference-in-Differences Settings with Staggered Adoption,” *Journal of Econometrics*, 226, 62–79.
- Basker, E. (2005), “Job Creation or Destruction? Labor Market Effects of Wal-Mart Expansion,” *Review of Economics and Statistics*, 87, 174–183.
- Belloni, A., Chernozhukov, V. and Hansen, C. (2014), “Inference on Treatment Effects after Selection among High-Dimensional Controls,” *Review of Economic Studies*, 81, 608–650.
- Borusyak, K., Jaravel, X. and Spiess, J. (2024), “Revisiting Event Study Designs: Robust and Efficient Estimation,” *Review of Economic Studies*, 91, 3253–3285.
- Brown, N. L. and Butts, K. (2025), “Dynamic Treatment Effect Estimation with Interactive Fixed Effects and Short Panels,” *Journal of Econometrics*, 250, 106013.
- Callaway, B. (2023), “Difference-in-Differences for Policy Evaluation,” in K. F. Zimmermann, ed., *Handbook of Labor, Human Resources and Population Economics*, Springer, Cham, 1–61.
- Callaway, B. and Sant’Anna, P. H. (2021), “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, 225, 200–230.

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. and Robins, J. (2018), “Double/debiased machine learning for treatment and structural parameters,” *The Econometrics Journal*, 21, C1–C68.
- de Chaisemartin, C. and d’Haultfoeuille, X. (2020), “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects,” *American Economic Review*, 110, 2964–2996.
- de Chaisemartin, C. and d’Haultfoeuille, X. (2023), “Two-Way Fixed Effects and Difference-in-Differences with Heterogeneous Treatment Effects: A Survey,” *Econometrics Journal*, 26, C1–C30.
- Goodman-Bacon, A. (2021), “Difference-in-Differences with Variation in Treatment Timing,” *Journal of Econometrics*, 225, 254–277.
- Harmon, N. A. (2024), “Difference-in-Differences and Efficient Estimation of Treatment Effects,” *Working Paper*.
- Heckman, J. J., Ichimura, H. and Todd, P. E. (1997), “Matching as an Econometric Evaluation Estimator: Evidence from Evaluating a Job Training Programme,” *Review of Economic Studies*, 64, 605–654.
- Hur, E. K., Lee, S. J. and Wooldridge, J. M. (2026), “Rolling Difference-in-Differences Estimation for Small and Large Panels,” *Available at SSRN 6502558*.
- Marcus, M. and Sant’Anna, P. H. (2021), “The Role of Parallel Trends in Event Study Settings: An Application to Environmental Economics,” *Journal of the Association of Environmental and Resource Economists*, 8, 235–275.
- Negi, A. and Wooldridge, J. M. (2021), “Revisiting Regression Adjustment in Experiments with Heterogeneous Treatment Effects,” *Econometric Reviews*, 40, 504–534.

- Neumark, D., Zhang, J. and Ciccarella, S. (2008), “The Effects of Wal-Mart on Local Labor Markets,” *Journal of Urban Economics*, 63, 405–430.
- Newey, W. K. and McFadden, D. (1994), “Large Sample Estimation and Hypothesis Testing,” in R. F. Engle and D. McFadden, eds, *Handbook of Econometrics*, Vol. 4, Elsevier, Amsterdam, 2111–2245.
- Rambachan, A. and Roth, J. (2023), “A More Credible Approach to Parallel Trends,” *Review of Economic Studies*, 90, 2555–2591.
- Sant’Anna, P. H. and Zhao, J. (2020), “Doubly Robust Difference-in-Differences Estimators,” *Journal of Econometrics*, 219, 101–122.
- Sun, L. and Abraham, S. (2021), “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects,” *Journal of Econometrics*, 225, 175–199.
- Wooldridge, J. M. (2007), “Inverse Probability Weighted Estimation for General Missing Data Problems,” *Journal of Econometrics*, 141, 1281–1301.
- Wooldridge, J. M. (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd edition, MIT Press, Cambridge, MA.
- Wooldridge, J. M. (2023), “Simple Approaches to Nonlinear Difference-in-Differences with Panel Data,” *Econometrics Journal*, 26, C31–C66.
- Wooldridge, J. M. (2025a), *Introductory Econometrics: A Modern Approach*, 8th edition, Cengage Learning, Boston, MA.
- Wooldridge, J. M. (2025b), “Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators,” *Empirical Economics*, 69, 2545–2587.

Appendices

A Proof of Theorem 3.1

We modify the argument in Wooldridge (2025b, Theorem 8.1). The estimators $\hat{\tau}_r$ are obtained from regression (3.3). Because $\dot{Y}_{ir} = Y_{ir} - \bar{Y}_{i,pre}$, basic OLS algebra shows that all coefficients from (3.3) can be obtained by differencing the coefficients from the following two regressions:

$$Y_{ir} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot \dot{\mathbf{X}}_i, i = 1, 2, \dots, N \quad (\text{A.1})$$

$$\bar{Y}_{i,pre} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot \dot{\mathbf{X}}_i, i = 1, 2, \dots, N \quad (\text{A.2})$$

where $\dot{\mathbf{X}}_i = \mathbf{X}_i - \bar{\mathbf{X}}_1$ are the covariates demeaned using the treated units. In particular, letting $\hat{\rho}_r$ be the coefficient on D_i from (A.1) and $\hat{\rho}_{pre}$ the coefficient on D_i from (A.2),

$$\hat{\tau}_r = \hat{\rho}_r - \hat{\rho}_{pre} \quad (\text{A.3})$$

Note also that the coefficients on the “moderating” terms, $D_i \cdot \dot{\mathbf{X}}_i$, are also obtained by differencing across the two regressions.

To show (A.3) is the same as the coefficient on $D_i \cdot fr_t$ in (3.6), first note that, by Wooldridge (2025b, Theorem 3.2), we can drop $(fq_t, fq_t \cdot \mathbf{X}_i)$ for $q < S$ without affecting the estimates. In other words, the $\tilde{\tau}_r$ are the coefficients on $D_i \cdot fr_t$ in the regression

$$\begin{aligned} Y_{it} \text{ on } & 1, \mathbf{X}_i, D_i, D_i \cdot \mathbf{X}_i, fS_t, \dots, fT_t, fS_t \cdot \mathbf{X}_i, \dots, fT_t \cdot \mathbf{X}_i \\ & D_i \cdot fS_t, \dots, D_i \cdot fT_t, D_i \cdot fS_t \cdot \dot{\mathbf{X}}_i, \dots, D_i \cdot fT_t \cdot \dot{\mathbf{X}}_i \end{aligned} \quad (\text{A.4})$$

Now, define $\mathbf{H}_i \equiv (1, \mathbf{X}_i, D_i, D_i \cdot \mathbf{X}_i)$, a $1 \times 2(K+1)$ vector, and

$$\mathbf{L}_{it} \equiv \left(fS_t, fS_t \cdot \mathbf{X}_i, D_i \cdot fS_t, D_i \cdot fS_t \cdot \dot{\mathbf{X}}_i, \dots, fT_t, fT_t \cdot \mathbf{X}_i, D_i \cdot fT_t \cdot \dot{\mathbf{X}}_i \right), \quad (\text{A.5})$$

a row vector with $2(T - S + 1)(K + 1)$ elements.

Note that for $q \neq r$, $(fq_t, fq_t \cdot \mathbf{X}_i, D_i \cdot fq_t, D_i \cdot fq_t \cdot \dot{\mathbf{X}}_i)$ and $(fr_t, fr_t \cdot \mathbf{X}_i, D_i \cdot fr_t, D_i \cdot fr_t \cdot \dot{\mathbf{X}}_i)$ are orthogonal in sample because $fq_t \cdot fr_t = 0$. The full set of regressors in (A.4) is simply $(\mathbf{H}_i, \mathbf{L}_{it})$. With $p_t = fS_t + \dots + fT_t$, the post-treatment period indicator, $(1 - p_t)fr_t = 0$, $r = S, S + 1, \dots, T$, which means $(1 - p_t)\mathbf{L}_{it} = \mathbf{0}$. Therefore, the objective function underlying the regression in (A.4) can be written as

$$\min_{\theta, \delta} \sum_{i=1}^N \sum_{t=1}^T (1 - p_t)(Y_{it} - \mathbf{H}_i \theta)^2 + \sum_{i=1}^N \sum_{t=1}^T p_t(Y_{it} - \mathbf{H}_i \theta - \mathbf{L}_{it} \delta)^2 \quad (\text{A.6})$$

Letting $\tilde{\theta}$ and $\tilde{\delta}$ denote the POLS estimators, the first-order conditions are

$$\sum_{i=1}^N \sum_{t=1}^T (1 - p_t) \mathbf{H}_i' (Y_{it} - \mathbf{H}_i \tilde{\theta}) + \sum_{i=1}^N \sum_{t=1}^T p_t \mathbf{H}_i' (Y_{it} - \mathbf{H}_i \tilde{\theta} - \mathbf{L}_{it} \tilde{\delta}) = \mathbf{0} \quad (\text{A.7})$$

$$\sum_{i=1}^N \sum_{t=1}^T p_t \mathbf{L}_{it}' (Y_{it} - \mathbf{H}_i \tilde{\theta} - \mathbf{L}_{it} \tilde{\delta}) = 0 \quad (\text{A.8})$$

Next, note that because $p_t = fS_t + \dots + fT_t$, we can write

$$p_t \mathbf{H}_i = \left[p_t, p_t \cdot D_i, p_t \cdot \mathbf{X}_i, p_t \cdot D_i \cdot \dot{\mathbf{X}}_i \right] = \sum_{q=S}^T \left[fq_t, fq_t \cdot D_i, fq_t \cdot \mathbf{X}_i, fq_t \cdot D_i \cdot \dot{\mathbf{X}}_i \right], \quad (\text{A.9})$$

which is simply the sum of the subvectors in $\mathbf{L}_{it}$ consisting of different time periods. It follows that $p_t \mathbf{H}_i = p_t \mathbf{L}_{it} \mathbf{A}$ for a $2(T - S + 1)(K + 1) \times 2(K + 1)$ matrix $\mathbf{A}$. Plugging into (A.8) gives

$$\sum_{i=1}^N \sum_{t=1}^T (1 - p_t) \mathbf{H}_i' (Y_{it} - \mathbf{H}_i \tilde{\theta}) + \mathbf{A}' \sum_{i=1}^N \sum_{t=1}^T p_t \mathbf{L}_{it}' (Y_{it} - \mathbf{H}_i \tilde{\theta} - \mathbf{L}_{it} \tilde{\delta}) = 0 \quad (\text{A.10})$$

Along with (A.8), (A.10) implies that the FOCs for $(\tilde{\theta}, \tilde{\delta})$ are

$$\sum_{i=1}^N \sum_{t=1}^T (1 - p_t) \mathbf{H}_i' (Y_{it} - \mathbf{H}_i \tilde{\theta}) = 0 \quad (\text{A.11})$$

But (A.11) means that $\tilde{\theta}$ is the OLS estimator from the regression Y_{it} on $\mathbf{H}_i$ using the pre-treatment period observations. With $\mathbf{H}_i$ not varying over time, $\tilde{\theta}$ is the same as the

cross-sectional regression

$$\bar{Y}_{i,pre} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot \dot{\mathbf{X}}_i \quad (\text{A.12})$$

In particular, the coefficient on D_i is precisely $\hat{\rho}_{pre}$ in (A.3).

Next, the FOC in (A.3) shows that $\tilde{\delta}$ is from a POLS regression using the post-treatment periods:

$$Y_{it} - \mathbf{H}_i \tilde{\theta} \text{ on } \mathbf{L}_{it}, t = S, \dots, T; i = 1, \dots, N$$

By definition of $\mathbf{L}_{it}$ and the orthogonality of the elements of $\mathbf{L}_{it}$ across the subvectors representing the different time periods, each $2(K+1)$ subvector, $\tilde{\delta}_r$, $r = S, \dots, T$, is obtained from a separate cross-sectional regression for each post-treatment period. Namely, because $fr_r = 1$, the regression for period r is

$$Y_{ir} - \mathbf{H}_i \tilde{\theta} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot \dot{\mathbf{X}}_i, i = 1, \dots, N \quad (\text{A.13})$$

The vector on right is simply $\mathbf{H}_i$, and so

$$\tilde{\delta}_r = \left(\sum_{i=1}^N \mathbf{H}_i' \mathbf{H}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{H}_i' Y_{ir} \right) - \tilde{\theta} \quad (\text{A.14})$$

The first term is the regression coefficients from

$$Y_{ir} \text{ on } 1, D_i, \mathbf{X}_i, D_i \cdot \dot{\mathbf{X}}_i, i = 1, \dots, N,$$

which is $\hat{\rho}_r$ from (A.3). We have shown that the coefficient corresponding to $D_i \cdot fr_t$ in the regression (A.4) is $\hat{\rho}_r - \hat{\rho}_{pre}$, which establishes the equivalence of the pooled OLS estimator across all time periods, (3.6), and the cross-sectional OLS estimators using the transformed variable $\dot{Y}_{ir}$ for each $r \in \{S, \dots, T\}$ separately. Essentially the same argument shows that the coefficients on the interaction terms $D_i \cdot fr_t \cdot \dot{\mathbf{X}}_i$ in (3.6) are the same as the coefficients on $D_i \cdot \dot{\mathbf{X}}_i$ in (3.3). $\square$

B Proof of the Detrending Procedure

For a cohort g , where we require $g \geq 3$ so there are at least two pre-treatment periods, define the $(g-1) \times 2$ matrix

$$\mathbf{J}_{g-1} = [\mathbf{1}_{g-1}, (1, 2, \dots, g-1)'] \quad (\text{B.1})$$

and let $\mathbf{Y}_{i,g-1}(\infty)$ be the $(g-1) \times 1$ vector: $\mathbf{Y}_{i,g-1}(\infty) = [Y_{i1}(\infty), \dots, Y_{i,g-1}(\infty)]'$.

A similar definition holds for $\mathbf{U}_{i,g-1}(\infty)$. Also, let $\mathbf{M}_{i,g-1}$ be the $(g-1) \times 1$ vector with elements $m_t(\mathbf{X}_i)$, $t = 1, \dots, g-1$. Note that we can write

$$\begin{aligned} \mathbf{Y}_{i,g-1}(\infty) &= \mathbf{J}_{g-1} \begin{pmatrix} \mathbf{h}(\mathbf{D}_i, \mathbf{X}_i) \\ \mathbf{D}_i \eta \end{pmatrix} + \mathbf{M}_{i,g-1} + \mathbf{U}_{i,g-1}(\infty) \\ &\equiv \mathbf{J}_{g-1} \mathbf{Q}_i + \mathbf{M}_{i,g-1} + \mathbf{U}_{i,g-1}(\infty) \end{aligned} \quad (\text{B.2})$$

Now regress $\mathbf{Y}_{i,g-1}(\infty)$ on $\mathbf{J}_{g-1}$, and obtain the coefficients,

$$\begin{aligned} \hat{\mathbf{B}}_{i,g-1} &= (\mathbf{J}_{g-1}' \mathbf{J}_{g-1})^{-1} \mathbf{J}_{g-1}' \mathbf{Y}_{i,g-1}(\infty) \\ &= \mathbf{Q}_i + (\mathbf{J}_{g-1}' \mathbf{J}_{g-1})^{-1} \mathbf{J}_{g-1}' [\mathbf{M}_{i,g-1} + \mathbf{U}_{i,g-1}(\infty)] \end{aligned} \quad (\text{B.3})$$

For $r \geq g$, the prediction of $Y_{ir}(\infty)$ using the unit-specific linear trend up through period $g-1$ is

$$\hat{Y}_{irg}(\infty) \equiv (1, r) \hat{\mathbf{B}}_{i,g-1} \quad (\text{B.4})$$

Use these predicted values to detrend $Y_{ir}(\infty)$:

$$\begin{aligned} \ddot{Y}_{irg}(\infty) &\equiv Y_{ir}(\infty) - \hat{Y}_{irg}(\infty) = Y_{ir}(\infty) - (1, r) \hat{\mathbf{B}}_{i,g-1} = (1, r) \mathbf{Q}_i + m_r(\mathbf{X}_i) + U_{ir}(\infty) \\ &\quad - (1, r) \left\{ \mathbf{Q}_i + (\mathbf{J}_{g-1}' \mathbf{J}_{g-1})^{-1} \mathbf{J}_{g-1}' [\mathbf{M}_{i,g-1} + \mathbf{U}_{i,g-1}(\infty)] \right\} \\ &= m_r(\mathbf{X}_i) + U_{ir}(\infty) - (1, r) \left\{ (\mathbf{J}_{g-1}' \mathbf{J}_{g-1})^{-1} \mathbf{J}_{g-1}' [\mathbf{M}_{i,g-1} + \mathbf{U}_{i,g-1}(\infty)] \right\} \end{aligned} \quad (\text{B.5})$$

The expression in (B.5) shows that $\ddot{Y}_{ir}(\infty)$ does not depend on $\mathbf{D}_i$. In particular,

$$\begin{aligned} E \left[\ddot{Y}_{irg}(\infty) | \mathbf{D}_i, \mathbf{X}_i \right] &= E \left[\ddot{Y}_{irg}(\infty) | \mathbf{X}_i \right] \\ &= m_r(\mathbf{X}_i) - (1, r) \left(\mathbf{J}'_{g-1} \mathbf{J}_{g-1} \right)^{-1} \mathbf{J}'_{g-1} \mathbf{M}_{i,g-1} \end{aligned} \quad (\text{B.6})$$

This conclusion is practically important because it shows that the vector of treatment cohort indicators, $\mathbf{D}_i$, are unconfounded with respect to the detrended variable $\ddot{Y}_{ir}(\infty)$, conditional on $\mathbf{X}_i$. Note how this extends the argument in Section 4 where, instead of $\mathbf{J}_{g-1}$ having rows $(1, t)$, its rows simply consisted of unity.

Now the modification to the arguments in Section 4 are straightforward. In place of (4.3) we have

$$Y_{ir}(g) - Y_{ir}(\infty) = \ddot{Y}_{irg}(g) - \ddot{Y}_{irg}(\infty) + \left[\hat{Y}_{irg}(g) - \hat{Y}_{irg}(\infty) \right], \quad (\text{B.7})$$

where $\ddot{Y}_{irg}(g) \equiv Y_{ir}(g) - \hat{Y}_{irg}(g)$ and $\hat{Y}_{irg}(g)$ are the predicted values from (B.5) and (B.4) but with $\mathbf{Y}_{i,g-1}(g)$ and $Y_{ir}(g)$ in place of $\mathbf{Y}_{i,g-1}(\infty)$ and $Y_{ir}(\infty)$. Now take the expectation conditional on $D_g = 1$:

$$\begin{aligned} \tau_{gr} &= E \left[Y_{ir}(g) - Y_{ir}(\infty) | D_{ig} = 1 \right] \\ &= E \left[\ddot{Y}_{irg}(g) - \ddot{Y}_{irg}(\infty) | D_{ig} = 1 \right] + E \left[\hat{Y}_{irg}(g) - \hat{Y}_{irg}(\infty) | D_{ig} = 1 \right] \end{aligned} \quad (\text{B.8})$$

The second term in (B.8) is zero by no anticipation because $\hat{Y}_{irg}(g)$ and $\hat{Y}_{irg}(\infty)$ are the same linear functions of the potential outcomes in periods $\{1, 2, \dots, g-1\}$. Therefore,

$$\tau_{gr} = E \left[\ddot{Y}_{irg}(g) - \ddot{Y}_{irg}(\infty) | D_{ig} = 1 \right] \quad (\text{B.9})$$

C Monte Carlo Simulations

In this supplementary section, we conduct Monte Carlo simulations for the common-timing case to study the finite-sample properties of our proposed estimators and compare them with competing approaches. We consider five estimators. The first is the POLS/ETWFE estimator of [Wooldridge \(2025b\)](#), which is efficient under a standard set of assumptions, although it is not doubly robust. In the present setting, the rolling RA estimator is numerically equivalent to POLS/ETWFE, and therefore the two are reported jointly as POLS/RA in the tables. We also consider the rolling IPWRA estimator and propensity score matching (PSM), both applied to the transformed outcome. Finally, we include the estimator of [Callaway and Sant’Anna \(2021\)](#), which is based on augmented inverse probability weighting (AIPW), another doubly robust approach. For [Callaway and Sant’Anna \(2021\)](#), we use the never-treated group as the control group and apply the transformation in equation (4.12).

We assume that Assumptions NAC and CPTC hold, along with overlap. However, we consider scenarios where the functional form of the conditional means and the functional form of the propensity score can be misspecified.

For each scenario, we use $T = 6$ with the first treatment in $S = 4$, which implies three post-treatment periods. Across Monte Carlo simulations we draw random samples of sizes 100, 500, and 1,000. We report bias, Monte Carlo standard deviation, and the root mean squared error (RMSE) of each estimator. All simulations use 1,000 Monte Carlo replications.

C.1 Data Generation.

We generate the data as follows. Two control variables are included: $\mathbf{X} = (X_1, X_2)$, where X_1 and X_2 are independent with $X_1 \sim \text{Gamma}(2, 2)$ [and so $E(X_1) = 4$] and $X_2 \sim \text{Bernoulli}(0.6)$.

The treatment indicator, D , has propensity score

$$p(\mathbf{x}) = P(D = 1 | \mathbf{X} = \mathbf{x}) = \frac{\exp(\mathbf{Z}_1 \gamma_1)}{1 + \exp(\mathbf{Z}_1 \gamma_1)} \quad (\text{C.1})$$

where the propensity score index function, $\mathbf{Z}_1 \gamma_1$, is

$$\mathbf{Z}_1 \gamma_1 = -1.2 + \frac{(X_1 - 4)}{2} - X_2 \quad (\text{C.2})$$

Our second step is generating heterogeneous treatment effects as follows:

$$\tau_r(\mathbf{X}) = \theta \cdot \sum_{q=S}^r (q - S + 1)^{-1} + \lambda_r \cdot h(\mathbf{X}), \quad r \in \{S, \dots, T\}, \quad (\text{C.3})$$

where $\theta = T - S + 1$ and λ_r is a time-varying parameter, set as $(\lambda_S, \dots, \lambda_T) = (0.5, 0.6, 1.0)$ in each simulation. This setup allows dynamic effects of being treated to vary across time and to increase as the length of exposure to the treatment increases. We consider two different functional forms of $h(\mathbf{X})$ in simulations. The first is

$$h(\mathbf{X}) = \frac{(X_1 - 4)}{2} + \frac{X_2}{3} \quad (\text{C.4})$$

In the second, $h(\mathbf{X})$ includes a quadratic in X_1 and an interaction between X_1 and X_2 :

$$h(\mathbf{X}) = \frac{(X_1 - 4)}{2} + \frac{X_2}{3} + \frac{(X_1 - 4)^2}{4} + (X_1 - 4) \cdot \frac{X_2}{2} \quad (\text{C.5})$$

We generate the potential outcomes in the untreated state as

$$Y_t(0) = \delta_t + C + \beta_t \cdot f(\mathbf{X}) + U_t(0), \quad (\text{C.6})$$

where $\delta_t = t$ is a time-specific component, $C|D, X \sim \text{Normal}(2, 1)$ is an individual-specific component, $U_t(0)|D, \mathbf{X} \sim \text{Normal}(0, 4)$ is the time-varying shock.

The time-varying β_t allows the effect of the covariates on potential outcome paths to

vary across time. For each simulation, the parameters are fixed as below:

$$\beta' = (\beta_1, \beta_2, \dots, \beta_T) = (1.0, 1.5, 0.8, 1.5, 2, 2.5) \quad (\text{C.7})$$

We consider two functional forms for $f(\mathbf{X})$:

$$f(\mathbf{X}) = \frac{(X_1 - 4)}{3} + \frac{X_2}{2} \quad (\text{C.8})$$

and

$$f(\mathbf{X}) = \frac{(X_1 - 4)}{3} + \frac{X_2}{2} + \frac{(X_1 - 4)^2}{3} + (X_1 - 4) \cdot \frac{X_2}{4} \quad (\text{C.9})$$

Finally, the post-treatment period outcome in the treated state is generated as

$$Y_t(1) = \begin{cases} Y_t(0), & t < S \\ Y_t(0) + \tau_t + U_t(1) - U_t(0), & t \geq S \end{cases}$$

where $U_t(1) | D, \mathbf{X} \sim \text{Normal}(0, 4)$.

For each simulation, all estimators that involve estimating the conditional means of Y_t assume the correct model is linear in $\mathbf{X}$. Therefore, when (C.4) and (C.8) are used in simulations, the conditional mean is correctly specified. However, when the data are generated as in (C.5) and (C.9), quadratic term X_1^2 and an interaction term $X_1 \cdot X_2$ are included; the conditional mean is misspecified.

We also consider a case where $(X_1 - 4)^2/2$ is added to the index function in the propensity score, and so the estimated logit model, which is always estimated assuming an index linear in X_1 and X_2 , is misspecified:

$$\mathbf{Z}_2\gamma_2 = -1.2 + \frac{(X_1 - 4)}{2} - X_2 + \frac{(X_1 - 4)^2}{2} \quad (\text{C.10})$$

Table A1 describes basic setups for each scenario.

Table A1: Scenarios with Common Timing

	Conditional Mean			Propensity Score	
	Correctly Specified?	$h(\mathbf{X})$	$f(\mathbf{X})$	Correctly Specified?	PS Index Function
Scenario 1C	Yes	(C.4)	(C.8)	Yes	(C.2)
Scenario 2C	Yes	(C.4)	(C.8)	No	(C.10)
Scenario 3C	No	(C.5)	(C.9)	Yes	(C.2)
Scenario 4C	No	(C.5)	(C.9)	No	(C.10)

C.2 Simulation Results

Table A2 reports results for Scenario 1C from Table A1, in which both the conditional mean and propensity score are correctly specified. As expected, all estimators exhibit little bias relative to the corresponding treatment effects.

Even when $N = 100$, the biases remain modest. Because the POLS/RA estimator is the best linear unbiased estimator in this setting, it is not surprising that it has smaller standard deviations than PSM and CS (2021).

Table A2: Correctly Specified Conditional Mean and Propensity Score

	N	Bias	τ_4 SD	RMSE	Bias	τ_5 SD	RMSE	Bias	τ_6 SD	RMSE
Sample ATT			3.205			4.746			5.909	
POLS/RA	100	0.001	0.664	0.664	-0.027	0.720	0.720	-0.044	0.703	0.704
PSM	100	-0.008	0.922	0.922	-0.025	0.946	0.947	-0.029	0.959	0.960
IPWRA	100	-0.016	0.693	0.693	-0.032	0.732	0.732	-0.054	0.722	0.724
CS(2021)	100	-0.004	0.865	0.865	-0.020	0.896	0.897	-0.033	0.891	0.891
Sample ATT			3.207			4.749			5.915	
POLS/RA	500	0.008	0.292	0.292	0.008	0.301	0.301	0.005	0.290	0.290
PSM	500	0.013	0.403	0.403	0.010	0.417	0.417	0.021	0.412	0.413
IPWRA	500	0.008	0.295	0.295	0.005	0.305	0.305	0.002	0.295	0.295
CS(2021)	500	0.018	0.371	0.372	0.016	0.378	0.378	0.013	0.367	0.367
Sample ATT			3.206			4.747			5.912	
POLS/RA	1,000	-0.003	0.203	0.203	-0.004	0.205	0.205	-0.011	0.210	0.210
PSM	1,000	-0.006	0.287	0.287	-0.002	0.287	0.287	-0.008	0.300	0.300
IPWRA	1,000	-0.003	0.209	0.209	-0.004	0.206	0.206	-0.014	0.215	0.215
CS(2021)	1,000	-0.007	0.255	0.255	-0.009	0.252	0.252	-0.018	0.262	0.262

Note 1: The population R -squared values are about 0.61, 0.60, and 0.60, respectively.

Note 2: The average propensity score is about 0.16.

When $N = 500$, the standard deviation of PSM for τ_4 is about 38 percent larger than that of POLS/RA, while the corresponding standard deviation for CS is about 27 percent

larger. By contrast, rolling IPWRA preserves nearly all of the efficiency of POLS/RA, with standard deviations that are at most about 2 percent larger.

In terms of RMSE, POLS/RA performs best throughout Table A2, which is again consistent with its BLUE property. Rolling IPWRA is a very close competitor, with only slightly larger RMSE values in most cases. For example, when $N = 1,000$, the RMSE of rolling IPWRA for τ_6 is 0.215, compared with 0.210 for POLS/RA.

Table A3 reports the findings for Scenario 3C, where the conditional mean is misspecified because the linear regression omits the terms X_1^2 and $X_1 \cdot X_2$. Because the propensity score is correctly specified in this scenario, estimators that rely on the propensity score, as well as doubly robust estimators, remain valid despite the misspecified conditional mean. By contrast, POLS/RA relies directly on the misspecified outcome regression and is therefore expected to exhibit systematic bias.

Table A3: Misspecified Conditional Mean and Correctly Specified Propensity Score

	N	Bias	τ_4 SD	RMSE	Bias	τ_5 SD	RMSE	Bias	τ_6 SD	RMSE
Sample ATT			3.510			5.112			6.520	
POLS/RA	100	0.053	0.727	0.729	0.090	0.861	0.866	0.138	0.996	1.006
PSM	100	0.021	0.961	0.961	0.042	1.059	1.060	0.075	1.192	1.195
IPWRA	100	-0.007	0.739	0.739	-0.011	0.840	0.840	-0.021	0.945	0.945
CS(2021)	100	0.000	0.946	0.946	-0.013	1.078	1.078	-0.024	1.187	1.188
Sample ATT			3.513			5.116			6.527	
POLS/RA	500	0.057	0.319	0.324	0.119	0.357	0.376	0.177	0.429	0.464
PSM	500	0.027	0.418	0.419	0.040	0.463	0.465	0.068	0.517	0.522
IPWRA	500	0.012	0.316	0.316	0.015	0.348	0.349	0.017	0.399	0.399
CS(2021)	500	0.020	0.407	0.407	0.019	0.447	0.448	0.018	0.511	0.511
Sample ATT			3.509			5.111			6.519	
POLS/RA	1,000	0.044	0.222	0.227	0.103	0.249	0.269	0.154	0.300	0.337
PSM	1,000	0.002	0.301	0.301	0.015	0.326	0.326	0.018	0.376	0.376
IPWRA	1,000	0.000	0.224	0.224	0.003	0.239	0.239	-0.003	0.282	0.282
CS(2021)	1,000	-0.003	0.275	0.275	-0.002	0.295	0.295	-0.009	0.344	0.345

Note 1: The population R -squared values are about 0.66, 0.65, and 0.65, respectively.

Note 2: The average propensity score is about 0.16.

In the simulations, POLS/RA indeed exhibits the largest bias, although the bias remains fairly small relative to the treatment effects. For example, when $N = 1,000$, the bias of POLS/RA for τ_6 is 0.154, compared with 0.018, -0.003, and -0.009 for PSM, rolling

IPWRA, and CS (2021), respectively. Even so, 0.154 remains small relative to the effect size of 6.519.

In terms of RMSE, rolling IPWRA performs best overall. It is uniformly the most efficient among the consistent estimators and attains the smallest RMSE in all but one design reported in Table A3. For example, for $\hat{\tau}_6$ with $N = 1,000$, the RMSE is smallest for rolling IPWRA at 0.282, followed by POLS/RA at 0.337, CS at 0.345, and PSM at 0.376. Thus, when the conditional mean is misspecified, applying IPWRA to the transformed outcome provides a clear robustness advantage: it substantially reduces bias relative to POLS/RA while also delivering greater finite-sample efficiency than CS.

D Pre-treatment Dynamics and Event Study Plot

This section describes how we construct transformed outcomes in the pre-treatment periods to estimate placebo effects and plot event-study dynamics.

D.1 Demeaning Transformation, $\dot{Y}_{itg}$

We apply a unit-specific demeaning transformation based on the full set of pre-treatment observations. Specifically, for each treated cohort g , we define

$$\dot{Y}_{itg} \equiv Y_{it} - \frac{1}{g-1} \sum_{q=1}^{g-1} Y_{iq}, \quad t = 1, \dots, T, \quad t \neq g-1. \quad (\text{D.1})$$

This transformation expresses outcomes relative to the unit-specific pre-treatment mean, which requires at least one pre-treatment period; $t = g-1$ is therefore excluded.

D.2 Detrending Transformation, $\ddot{Y}_{itg}$

For detrending, we remove a unit-specific linear trend estimated using the full set of pre-treatment observations. For each treated cohort g , let

$$\hat{Y}_{itg} = \hat{a}_{ig} + \hat{b}_{ig}t$$

denote the fitted value obtained from regressing Y_{iq} on q using all pre-treatment periods $q < g$. The detrended outcome is then defined as

$$\ddot{Y}_{itg} = Y_{it} - \hat{Y}_{itg}, \quad t = 1, \dots, T, \quad t \notin \{g-2, g-1\}. \quad (\text{D.2})$$

This transformation expresses each outcome as a deviation from the fitted unit-specific pre-treatment trend, whose level and slope are estimated from the pre-treatment periods; as this requires at least two pre-treatment periods, $t = g-2$ and $t = g-1$ are therefore excluded.

D.3 Estimation

Once the transformed outcome variables are constructed, the estimation procedure follows the same steps as in the main text.

For identification, the comparison sample for $ATT(g, t)$ is defined dynamically as a function of both cohort g and calendar time t . Specifically, for each cohort–time cell (g, t) , we compare units first treated in period g with units that are either never treated or treated strictly after $\max(g, t)$. Formally,

$$\mathcal{A}_{g,t} = \{i : G_i = g\} \cup \{i : G_i = 0\} \cup \{i : G_i > \max(g, t)\},$$

where G_i denotes the first treatment period, with $G_i = 0$ for never-treated units.

Thus, the comparison group differs across pre- and post-treatment periods. In pre-treatment periods ($t < g$), the control group consists of never-treated units and cohorts treated after g , so that

$$\mathcal{A}_{g,t} = \{i : G_i = g\} \cup \{i : G_i = 0\} \cup \{i : G_i > g\}.$$

In post-treatment periods ($t \geq g$), the control group consists of never-treated units and cohorts that remain untreated at time t , so that

$$\mathcal{A}_{g,t} = \{i : G_i = g\} \cup \{i : G_i = 0\} \cup \{i : G_i > t\}.$$

We use the same cohort-specific comparison rule in pre-treatment periods to construct placebo estimates.

We then aggregate these cohort–time estimates across all available (g, t) cells with common relative time $r = t - g$ to obtain the event-time effect $WATT(r)$. Specifically,

$$WATT(r) = \sum_{g \in \mathcal{G}_r} \frac{N_g}{N_{\mathcal{G}_r}} ATT(g, g + r),$$

where N_g denotes the number of units in cohort g , and $N_{\mathcal{G}_r} = \sum_{h \in \mathcal{G}_r} N_h$. Thus, $WATT(r)$ is a weighted average of cohort-specific treatment effects, with weights proportional to cohort sizes.

Graphical Presentation The event-study graph displays the estimated $WATT(r)$ over relative time, together with the associated confidence bands. For $r \geq 0$, the estimates summarize immediate and dynamic post-treatment effects. For $r < 0$, the estimates provide pre-treatment diagnostics for systematic differences between treated and comparison units.

The interpretation of the pre-treatment estimates depends on the transformation. Under demeaning, the estimates measure treatment-control differences in unit-specific demeaned outcomes; once unit-specific level differences are removed, parallel trends implies that treated and comparison units follow the same pre-treatment path absent treatment, so the pre-treatment estimates are expected to be centered at zero.

Under detrending, the estimates measure treatment-control differences in deviations from fitted unit-specific linear trends; hence, the pre-treatment estimates diagnose whether differential dynamics remain after removing unit-specific trends.

In practice, the user-written Stata command `lwdid` implements these steps automatically, including the construction of event-study estimates and the generation of the corresponding event-study graph.

E Inference via Multiplier Bootstrap

Inference for the weighted average treatment effect on the treated, $WATT(r)$, is conducted using a multiplier bootstrap based on observation-level influence functions. To implement this approach, we compute observation-level influence functions for each group–time $ATT(g, t)$ estimator. Importantly, we use the *influence functions* of the estimators.

These influence contributions provide an asymptotic linear representation for $\widehat{WATT}(r)$ and allow us to approximate its sampling distribution without re-estimating the model in each bootstrap replication, substantially reducing computational cost. The bootstrap is applied to the event-time coefficients that are estimated and reported, excluding the anchor periods that are set to zero by construction.

We begin by outlining the aggregation and bootstrap steps used for inference on $WATT(r)$, and then turn to the influence-function derivations for the estimators within the rolling-transformation framework.

E.1 Algorithm Details

Let $\widehat{ATT}(g, t)$ denote the estimator of $ATT(g, t)$ obtained from one of the estimators considered in the paper (RA, IPW, or IPWRA). For each (g, t) cell, the estimator admits the asymptotic linear representation

$$\widehat{ATT}(g, t) - ATT(g, t) = \frac{1}{n_{g,t}} \sum_{i \in \mathcal{A}_{g,t}} IF_{i,g,t} + o_p(n_{g,t}^{-1/2}),$$

where $IF_{i,g,t}$ denotes the influence function contribution of observation i for the (g, t) cell. Its form depends on the estimator: for RA (Section E.2), it reduces to the OLS influence function, whereas for IPW and IPWRA (Sections E.4 and E.3), it also includes score corrections from first-stage propensity score estimation.

For inference on $WATT(r)$, define the aggregated influence function

$$IF_{i,r} = \sum_g \omega_{g,r} IF_{i,g,g+r},$$

where $\omega_{g,r}$ denotes the weight assigned to cohort g in the construction of $WATT(r)$, equal to the number of treated units in cohort g contributing at event time r divided by the total number of treated units contributing at that event time.

To improve finite-sample stability, the aggregated influence contributions are centered across observations,

$$\widetilde{IF}_{i,r} = IF_{i,r} - \frac{1}{N} \sum_{j=1}^N IF_{j,r}.$$

The multiplier bootstrap proceeds by drawing independent Rademacher multipliers $\xi_i^{(b)} \in \{-1, +1\}$ for each unit i , and constructing bootstrap draws of the estimation error

$$\widehat{WATT}^{*(b)}(r) = \sum_{i=1}^N \xi_i^{(b)} \widetilde{IF}_{i,r}.$$

Because the influence functions already contain the appropriate sample-size scaling, no additional normalization is required.

Bootstrap standard errors, denoted $\widehat{se}_{boot}(r)$, are obtained from the sample variance of $\{\widehat{WATT}^{*(b)}(r)\}_{b=1}^B$. By default, inference is based on simultaneous confidence bands that ensure joint coverage over all event times. For each bootstrap repetition b , define the supremum statistic

$$T_b^* = \sup_{r \in \mathcal{R}} \left| \frac{\widehat{WATT}^{*(b)}(r)}{\widehat{se}_{boot}(r)} \right|.$$

Let $\hat{c}_{1-\alpha}^{\sup}$ denote the empirical $(1 - \alpha)$ -quantile of $\{T_b^*\}_{b=1}^B$. The resulting simultaneous confidence band is

$$\widehat{WATT}(r) \pm \hat{c}_{1-\alpha}^{\sup} \widehat{se}_{boot}(r), \quad r \in \mathcal{R}.$$

Algorithm 1 Multiplier Bootstrap for Simultaneous Inference on $\widehat{WATT}(r)$

Require: Observation-level influence-function contributions $\widehat{IF}_{i,g,t}$, aggregation weights $\omega_{g,r}$, confidence level $1 - \alpha$, and number of bootstrap repetitions B .

Ensure: Bootstrap standard errors and simultaneous confidence bands for $\widehat{WATT}(r)$.

1. For each group–time cell (g, t) , compute $\widehat{ATT}(g, t)$ and the corresponding observation-level influence contributions $\widehat{IF}_{i,g,t}$.
2. Aggregate treatment effects across cohorts:

$$\widehat{WATT}(r) = \sum_g \omega_{g,r} \widehat{ATT}(g, g+r).$$

3. Construct event-time influence contributions:

$$\widehat{IF}_{i,r} = \sum_g \omega_{g,r} \widehat{IF}_{i,g,g+r},$$

$$\widetilde{IF}_{i,r} = \widehat{IF}_{i,r} - \frac{1}{N} \sum_{j=1}^N \widehat{IF}_{j,r}.$$

4. For each bootstrap repetition $b = 1, \dots, B$, draw independent unit-level Rademacher multipliers $\xi_i^{(b)} \in \{-1, +1\}$ and compute

$$\widehat{WATT}^{*(b)}(r) = \sum_{i=1}^N \xi_i^{(b)} \widetilde{IF}_{i,r}.$$

5. Compute bootstrap standard errors $\widehat{se}_{boot}(r)$ from the empirical variance of

$$\{\widehat{WATT}^{*(b)}(r)\}_{b=1}^B.$$

6. For each repetition b , define the supremum statistic

$$T_b^* = \sup_{r \in \mathcal{R}} \left| \frac{\widehat{WATT}^{*(b)}(r)}{\widehat{se}_{boot}(r)} \right|.$$

7. Let $\hat{c}_{1-\alpha}^{\sup}$ denote the empirical $(1 - \alpha)$ -quantile of $\{T_b^*\}_{b=1}^B$. The simultaneous confidence band is given by

$$\widehat{WATT}(r) \pm \hat{c}_{1-\alpha}^{\sup} \widehat{se}_{boot}(r), \quad r \in \mathcal{R}.$$

E.2 Influence Function for the RA Estimator

For a given cohort-period cell (g, t) , let $\dot{Y}_{i,g,t}$ denote the rolling-residualized outcome and let $D_{i,g} \in \{0, 1\}$ be the cohort indicator. Define the comparison sample for cell (g, t) as

$$\mathcal{A}_{g,t} = \{i : i \text{ belongs to cohort } g \text{ or is in the control group for } (g, t)\} \cap \{i : T_i = t\},$$

and let $n_{g,t} = |\mathcal{A}_{g,t}|$.

RA estimator with covariates We estimate $ATT(g, t)$ via the following regression-adjustment specification estimated by OLS on the sample $\mathcal{A}_{g,t}$:

$$\dot{Y}_{i,g,t} = \alpha_{g,t} + \theta_{g,t}D_{i,g} + X_i'\gamma_{g,t} + D_{i,g}(X_i - \bar{X}_g)'\delta_{g,t} + \varepsilon_{i,g,t}, \quad (\text{E.1})$$

where $\bar{X}_g = n_g^{-1} \sum_{i:G_i=g} X_i$ denotes the sample mean of covariates for treated units in cohort g .

Define the design vector

$$Z_i = (1, D_{i,g}, X_i', D_{i,g}(X_i - \bar{X}_g)')'.$$

Let

$$b_{g,t} = (\alpha_{g,t}, \theta_{g,t}, \gamma_{g,t}', \delta_{g,t}')'$$

denote the coefficient vector in (E.1). Stacking observations over $i \in \mathcal{A}_{g,t}$ yields

$$\hat{b}_{g,t} = (Z'Z)^{-1}Z'\dot{Y},$$

where Z and $\dot{Y}$ collect the rows Z_i and $\dot{Y}_{i,g,t}$, respectively. The ATT estimator is the coefficient on the treatment indicator,

$$\hat{\theta}_{g,t} = e_2'\hat{b}_{g,t},$$

where e_2 selects the coefficient on $D_{i,g}$. The interaction term $D_{i,g}(X_i - \bar{X}_g)$ allows slopes on X_i to differ across treated and control units, while centering at $\bar{X}_g$ ensures that $\hat{\theta}_{g,t}$ recovers the average treatment effect for the treated.

Influence function Because the estimator is OLS, its influence function has a closed-form expression. Let

$$\hat{u}_{i,g,t} = \dot{Y}_{i,g,t} - Z_i' \hat{b}_{g,t}$$

denote the OLS residual. Then the finite-sample influence contribution of $\hat{\theta}_{g,t}$ is

$$\widehat{IF}_{i,g,t}^{RA} = e_2'(Z'Z)^{-1} Z_i \hat{u}_{i,g,t}, \quad i \in \mathcal{A}_{g,t}. \quad (\text{E.2})$$

This follows directly from the OLS representation and therefore holds exactly in finite samples.

E.3 Influence Function for the IPWRA Estimator

The IPWRA estimator combines inverse probability weighting with regression adjustment. In our implementation, we follow the two-step procedure described in [Wooldridge \(2025a\)](#). Its influence function is derived from the stacked moment conditions of the estimator.

Setup Using the same notation as in Section [E.2](#), let $D_{i,g} = \mathbf{1}\{G_i = g\}$ denote the cohort indicator, $\dot{Y}_{i,g,t}$ the rolling-residualized outcome, X_i the covariate vector, and $\mathcal{A}_{g,t}$ the analysis sample used to estimate $ATT(g, t)$. Define $\bar{X}_g = n_g^{-1} \sum_{i:G_i=g} X_i$.

Propensity score The propensity score is defined as

$$p_{g,t}(X_i; \gamma_{g,t}) = \Pr(D_{i,g} = 1 \mid X_i, i \in \mathcal{A}_{g,t}) = \Lambda(X_i' \gamma_{g,t}),$$

where $\Lambda(u) = \frac{e^u}{1+e^u}$. The estimator $\hat{\gamma}_{g,t}$ is obtained by maximum likelihood.

Define the inverse probability weights

$$w_{i,g,t}(\gamma) = D_{i,g} + (1 - D_{i,g}) \frac{p_{g,t}(X_i; \gamma)}{1 - p_{g,t}(X_i; \gamma)}.$$

Weighted outcome regression Define the augmented regressor vector

$$Z'_{i,g} = (1, D_{i,g}, X'_i, D_{i,g}(X_i - \bar{X}_g)').$$

For each (g, t) cell, the IPWRA estimator is obtained from the following weighted least squares problem:

$$\hat{\eta}_{g,t} = \begin{pmatrix} \hat{\alpha}_{g,t} \\ \hat{\theta}_{g,t} \\ \hat{\beta}_{g,t} \\ \hat{\delta}_{g,t} \end{pmatrix} = \arg \min_{\eta} \sum_{i \in \mathcal{A}_{g,t}} w_{i,g,t}(\hat{\gamma}_{g,t}) (\dot{Y}_{i,g,t} - Z'_{i,g} \eta)^2.$$

The ATT estimator corresponds to the second component of $\eta_{g,t}$,

$$\hat{\theta}_{g,t} = e'_2 \hat{\eta}_{g,t}.$$

Stacked moment representation To derive the asymptotic distribution, we express the IPWRA estimator as a stacked M-estimator. Define the nuisance parameter vector

$$\vartheta_{g,t} = \begin{pmatrix} \eta_{g,t} \\ \gamma_{g,t} \end{pmatrix}.$$

Let the stacked moment function be

$$\Psi_{i,g,t}(\vartheta) = \begin{pmatrix} \psi_{i,g,t}^{\eta}(\eta, \gamma) \\ \psi_{i,g,t}^{\gamma}(\gamma) \end{pmatrix},$$

where

$$\psi_{i,g,t}^{\eta}(\eta, \gamma) = w_{i,g,t}(\gamma) Z_{i,g} (\dot{Y}_{i,g,t} - Z'_{i,g} \eta)$$

and $\psi_{i,g,t}^\gamma(\gamma)$ denotes the score of the logit likelihood used to estimate $\gamma_{g,t}$.

The estimator solves

$$\frac{1}{n_{g,t}} \sum_{i \in \mathcal{A}_{g,t}} \Psi_{i,g,t}(\hat{\vartheta}_{g,t}) = 0.$$

Influence function Under standard regularity conditions for M-estimators,

$$\sqrt{n_{g,t}}(\hat{\vartheta}_{g,t} - \vartheta_{g,t}) = \frac{1}{\sqrt{n_{g,t}}} \sum_{i \in \mathcal{A}_{g,t}} IF_{i,g,t}^\vartheta + o_p(1),$$

where

$$IF_{i,g,t}^\vartheta = -A_{g,t}^{-1} \Psi_{i,g,t}(\vartheta_{g,t}).$$

Partitioning the Jacobian matrix gives

$$A_{g,t} = \begin{pmatrix} Q_{w,g,t} & H_{g,t} \\ 0 & A_{\gamma,g,t} \end{pmatrix},$$

where

$$Q_{w,g,t} = E[w_{i,g,t} Z_{i,g} Z'_{i,g}], \quad H_{g,t} = E \left[Z_{i,g} \frac{\partial w_{i,g,t}}{\partial \gamma'} (\dot{Y}_{i,g,t} - Z'_{i,g} \eta_{g,t}) \right].$$

Let

$$u_{i,g,t} = \dot{Y}_{i,g,t} - Z'_{i,g} \eta_{g,t}.$$

The influence function for $\eta_{g,t}$ is

$$IF_{\eta,i,g,t} = Q_{w,g,t}^{-1} (w_{i,g,t} Z_{i,g} u_{i,g,t} + H_{g,t} IF_{\gamma,i,g,t}).$$

Since $ATT(g, t) = e'_2 \eta_{g,t}$, the influence function for the treatment effect estimator is

$$IF_{i,g,t}^{ATT} = e'_2 IF_{\eta,i,g,t}.$$

E.4 Influence Function for the IPW Estimator

As a special case of the IPWRA framework, the IPW estimator is obtained by omitting the outcome regression component and retaining only inverse probability weighting. Using the same notation as in Section E.3, the parameter vector reduces to $\vartheta_{g,t} = (\theta_{g,t}, \gamma_{g,t})$, and the moment system consists of the IPW estimating equation together with the logit score for $\gamma_{g,t}$.

The IPW estimating equation is

$$\psi_{i,g,t}^{IPW}(\theta, \gamma) = \frac{D_{i,g}}{E[D_{i,g}]}(\dot{Y}_{i,g,t} - \theta) - \frac{(1 - D_{i,g})p_{g,t}(X_i; \gamma)}{(1 - p_{g,t}(X_i; \gamma))E[D_{i,g}]} \dot{Y}_{i,g,t}.$$

Influence function Under standard regularity conditions, the estimator admits an asymptotic linear representation

$$\hat{\theta}_{g,t} - \theta_{g,t} = \frac{1}{n_{g,t}} \sum_{i \in \mathcal{A}_{g,t}} IF_{i,g,t}^{IPW} + o_p(n_{g,t}^{-1/2}).$$

Let

$$\psi_{i,g,t} = \frac{D_{i,g}}{E[D_{i,g}]}(\dot{Y}_{i,g,t} - \theta_{g,t}) - \frac{(1 - D_{i,g})p_{g,t}(X_i; \gamma_{g,t})}{(1 - p_{g,t}(X_i; \gamma_{g,t}))E[D_{i,g}]} \dot{Y}_{i,g,t}$$

denote the plug-in contribution, and let

$$IF_{\gamma,i,g,t} = A_{g,t}^{-1} X_i (D_{i,g} - p_{g,t}(X_i; \gamma_{g,t})), \quad A_{g,t} = E[X_i X_i' p_{g,t}(X_i; \gamma_{g,t}) (1 - p_{g,t}(X_i; \gamma_{g,t}))].$$

Define

$$\Gamma_{g,t} = E \left[X_i \frac{(1 - D_{i,g}) \dot{Y}_{i,g,t} p_{g,t}(X_i; \gamma_{g,t})}{1 - p_{g,t}(X_i; \gamma_{g,t})} \right].$$

Then the influence function reduces to

$$IF_{i,g,t}^{IPW} = \psi_{i,g,t} - \Gamma'_{g,t} IF_{\gamma,i,g,t}.$$

The second term captures the effect of first-stage propensity score estimation.

F Estimation Results

This section provides supplementary results related to our main analysis. Figure A1 depicts the distribution of counties by their first Walmart opening year: darker shades indicate later openings, the lightest green denotes never-treated counties (31% of the sample), and gray marks excluded areas. The treatment cohort is defined by the year of first opening.

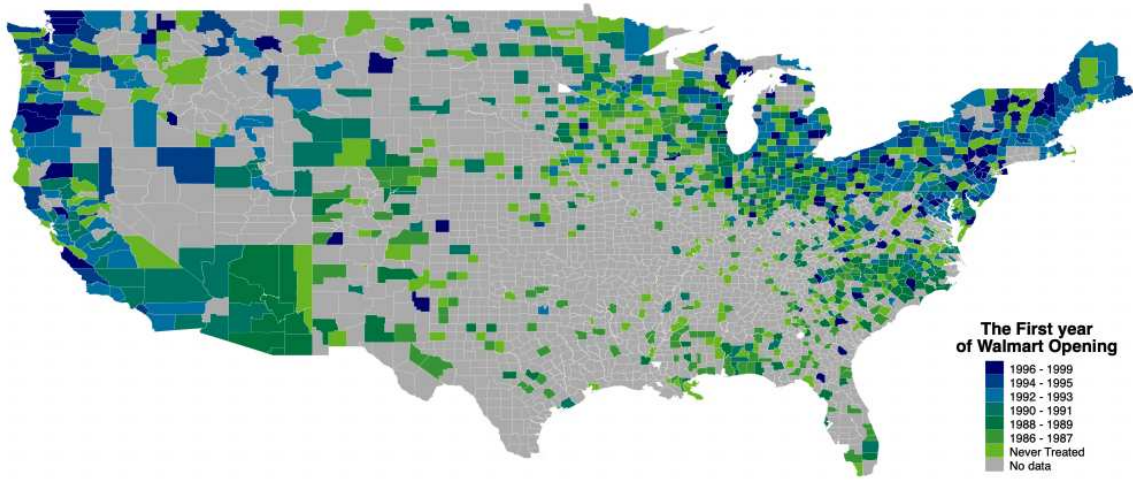


Figure A1: First Year of Walmart Store Openings by County

Table A4 presents estimation results, corresponding to the visual representation in Figure 1; the retail labor market case. We estimate standard errors of the weighted ATTs using the multiplier bootstrap of Appendix E with 999 repetitions. The first three columns display estimates of weighted ATTs from the ETWFE (Wooldridge, 2025b), CS (2021), and our rolling IPWRA estimator without unit-specific detrending, each of which is biased when the parallel trends assumption is violated. The fourth and fifth columns present the point estimates from our rolling RA and rolling IPWRA estimators, which appropriately account for potential heterogeneous linear trends.

Table A4: Effects of Walmart Openings on log(Retail employment)

r	ETWFE		CS (2021)		Rolling IPWRA		Heterogeneous Trend			
	ATT(r)	SE	ATT(r)	SE	ATT(r)	SE	Rolling RA		Rolling IPWRA	
	ATT(r)	SE	ATT(r)	SE	ATT(r)	SE	ATT(r)	SE	ATT(r)	SE
0	0.041	(0.006)	0.022	(0.003)	0.040	(0.006)	0.007	(0.004)	0.007	(0.004)
1	0.072	(0.007)	0.052	(0.004)	0.072	(0.007)	0.032	(0.005)	0.032	(0.005)
2	0.072	(0.008)	0.051	(0.004)	0.072	(0.008)	0.024	(0.006)	0.024	(0.006)
3	0.073	(0.009)	0.051	(0.005)	0.074	(0.009)	0.018	(0.008)	0.019	(0.008)
4	0.078	(0.010)	0.055	(0.006)	0.079	(0.010)	0.015	(0.009)	0.016	(0.009)
5	0.088	(0.011)	0.061	(0.007)	0.091	(0.011)	0.016	(0.011)	0.016	(0.011)
6	0.098	(0.012)	0.071	(0.009)	0.101	(0.012)	0.020	(0.013)	0.017	(0.013)
7	0.116	(0.013)	0.089	(0.010)	0.122	(0.013)	0.036	(0.015)	0.033	(0.015)
8	0.129	(0.014)	0.104	(0.011)	0.137	(0.015)	0.040	(0.017)	0.039	(0.016)
9	0.134	(0.016)	0.114	(0.013)	0.144	(0.017)	0.038	(0.021)	0.039	(0.019)
10	0.155	(0.019)	0.131	(0.015)	0.168	(0.019)	0.024	(0.025)	0.035	(0.022)
11	0.163	(0.023)	0.141	(0.019)	0.178	(0.023)	0.000	(0.031)	0.016	(0.027)
12	0.164	(0.030)	0.150	(0.024)	0.182	(0.029)	-0.002	(0.039)	0.015	(0.034)
13	0.204	(0.043)	0.188	(0.035)	0.231	(0.042)	0.028	(0.055)	0.046	(0.043)

F.1 Wholesale Employment

In this section, we report results for county-level wholesale employment. Figure A2 compares estimates from the CS (2021) approach with those from our rolling IPWRA estimators, with unit-specific demeaning and with unit-specific detrending.

Panels (a) and (b) of Figure A2 mirror the retail employment results: both the CS estimator and the rolling IPWRA estimator without detrending display upward pre-treatment trends.

By contrast, Panel (c), which applies the rolling IPWRA estimator with county-specific detrending, shows a substantially flatter pre-treatment path, with no statistically significant pre-treatment estimates. It suggests that the detrended specification effectively addresses differential pre-treatment dynamics between treated and comparison counties. The resulting negative estimates represent a sign reversal relative to the CS and demeaned specifications, and are consistent with Brown and Butts (2025), who use factor models to relax the parallel trends assumption.

Overall, the comparison highlights the importance of accounting for differential pre-treatment dynamics in this application.

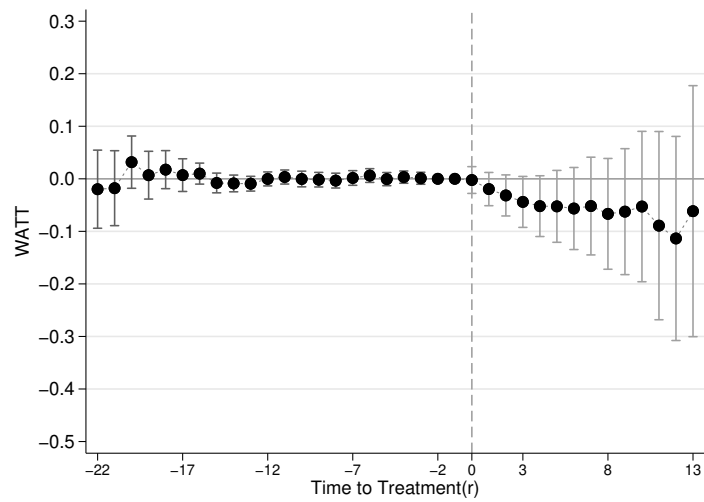
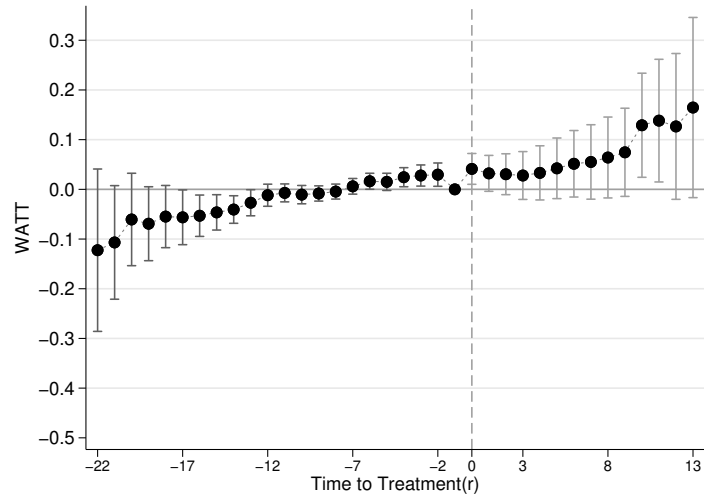
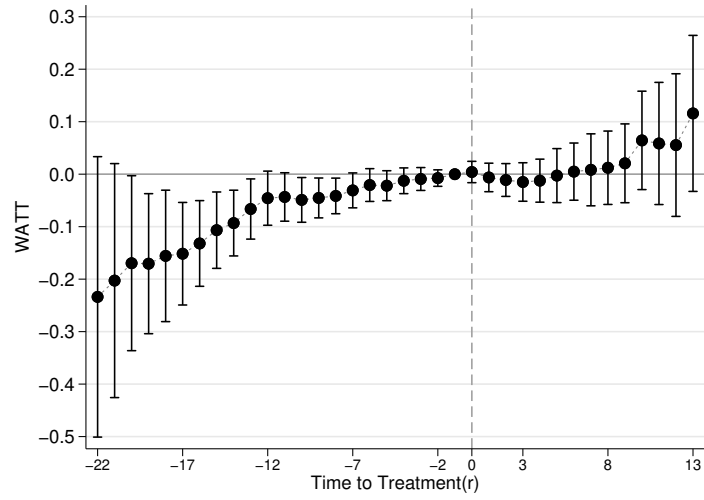


Figure A2: Effects of Walmart Openings on $\log(\text{Wholesale Employment})$

Table A5 shows estimates in line with Figure A2.

Table A5: Effects of Walmart Openings on log(Wholesale employment)

r	ETWFE		CS (2021)		Rolling IPWRA		Heterogeneous Trend			
	ATT(r)	SE	ATT(r)	SE	ATT(r)	SE	Rolling RA		Rolling IPWRA	
0	0.043	(0.011)	0.004	(0.007)	0.041	(0.010)	-0.001	(0.008)	-0.002	(0.008)
1	0.034	(0.012)	-0.006	(0.009)	0.032	(0.012)	-0.017	(0.010)	-0.020	(0.011)
2	0.031	(0.014)	-0.011	(0.010)	0.030	(0.014)	-0.028	(0.013)	-0.032	(0.013)
3	0.028	(0.016)	-0.015	(0.012)	0.028	(0.017)	-0.039	(0.016)	-0.044	(0.016)
4	0.032	(0.018)	-0.012	(0.014)	0.033	(0.019)	-0.046	(0.019)	-0.052	(0.020)
5	0.041	(0.020)	-0.003	(0.017)	0.042	(0.021)	-0.045	(0.023)	-0.052	(0.023)
6	0.048	(0.022)	0.005	(0.018)	0.051	(0.023)	-0.043	(0.026)	-0.057	(0.027)
7	0.049	(0.024)	0.008	(0.023)	0.055	(0.025)	-0.040	(0.031)	-0.052	(0.032)
8	0.054	(0.026)	0.012	(0.023)	0.064	(0.028)	-0.055	(0.035)	-0.067	(0.036)
9	0.061	(0.029)	0.021	(0.025)	0.075	(0.031)	-0.058	(0.040)	-0.062	(0.041)
10	0.111	(0.034)	0.064	(0.031)	0.129	(0.035)	-0.054	(0.047)	-0.053	(0.049)
11	0.117	(0.040)	0.059	(0.039)	0.138	(0.040)	-0.101	(0.057)	-0.089	(0.061)
12	0.100	(0.049)	0.055	(0.045)	0.127	(0.048)	-0.135	(0.067)	-0.114	(0.068)
13	0.134	(0.062)	0.116	(0.050)	0.165	(0.060)	-0.088	(0.083)	-0.062	(0.083)